The Membrane: Cognitive Boundary Architecture

A nervous-system firewall against AI routing and thought surveillance

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The Membrane: Cognitive Boundary Architecture

> **Independent research** by Zorie R. Barber — not affiliated with any company, employer, or organization. Third-party names cited in this document are descriptive only.

Abstract

The Membrane is a research architecture for a **cognitive boundary** — a firewall between endogenous human cognition (native nervous-system processing) and exogenous channels that can read, route, inject, or substitute for thought. The threat surface includes **AI routing** (cognition mediated through corporate inference endpoints, agents, and memory layers), **invasive neural read channels** (implants and BCIs), and **non-**

invasive inference (EEG, behavioral phenotyping, multimodal prediction). The design uses zk-STARK Chain Proofs, hardware attestation, and web-of-trust witnesses to detect unauthorized boundary crossings, fail closed when attestation breaks, and preserve sovereignty over what may cross the membrane. It does not prove consciousness or block a perfect covert read; it raises the cost of silent compromise and gives the subject a severance path.

Executive Summary

What this is: A specification for a human **cognition / thinking / nervous-system firewall**. The Membrane sits at the boundary where endogenous neural activity meets external systems — AI routers, cloud inference, BCI telemetry, and surveillance pipelines that reconstruct mental state from behavior or physiology.

Primary threats:

Class	Example
AI routing	Assistants, copilots, and agent frameworks that ingest context, propose completions, and shape decisions through off-device models
Invasive read / write	Cortical implants, consumer BCIs, and medical devices with upstream telemetry or stimulation channels
Non-invasive inference	EEG/MEG/fNIRS, gaze + keystroke models, emotion classifiers, digital phenotyping, predictive policing of intent
Outsourced wetware compute	Code-deployable biological computers and biological cloud APIs coupling external neuron cultures in closed read/write loops
BioLLM-class hybrid	LLM token routing modulated by living neurons on API (wetware bias inside router session) — Class A + Class H in one loop[1][2]

Mechanism: Recursive zk-STARK Chain Proofs bind (1) a neural/biometric anchor, (2) periodic **cognitive-integrity signals** from approved channels, (3) **Intent Authorization Certificates** before decode or router action, (4) witness corroboration, and (5) TEE-backed node attestation. If domains disagree or a channel lacks valid attestation, the membrane **severs** — blocking routed inference, closing BCI sessions, or degrading to offline-safe mode.

What this is NOT: - Proof of consciousness or qualia. - Proof that no off-chain clone or hidden reader exists. - A shipping product. Phase 0 is a research prototype. - The work, opinion, or roadmap of any company or institution named in this document.

Major limitations: - Covert surveillance with no attested channel may be undetectable. - TEEs and BCIs are best-available anchors, not absolute roots of trust. - Cognitive-integrity signals are probabilistic; replay, injection, and coercion remain. - Closed-loop BCI or wetware coupling induces **minute-scale neural drift**; liveness commitments must bind task, decoder, and stimulation policy — not static biometrics alone.[3] - Fork detection is chain-bound; hidden parallel readers are out of scope.

Data structures: SHA-256 Merkle trees for channel features, bus events, witness sets, and periodic CP rollups; OpenTimestamps cold anchor on daily rollup (§5.1, §11.4).

Phase 0 prototype: Self-hosted attestation bus, TEE node, `Liveness-2`-ready neural/sensor commitment circuit, and K=2–3 personal witnesses. See §9.

Future extensions: Full BCI integration (§4.6), agent-handshake mode for untrusted AI delegates (§4.7), GPU-accelerated proving (§10.1).

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Glossary

Term	Definition
CP	Chain Proof. A zk-STARK attestation that binds cognitive anchor, channel integrity, witness set, and node state into one recursive proof.
RFA	Recursive Attestation. Each CP commits to the node's previous attestation, forming an audit chain.
T_0	Cognitive anchor. One-time T_0 enrollment binding a human to cryptographic identity via neural/biometric commitment and social witnesses.
Cognitive-integrity signal	Time-bounded evidence that attested channels (BCI, local TEE, approved routers) match expected state. Not proof of thought content.
Membrane	The boundary layer between endogenous cognition and exogenous read/write/routing channels.
AI routing	Mediation of cognition through external inference systems (LLMs, agents, cloud memory) that ingest context and return outputs.
Liveness-2	Neural-channel circuit: BCI-derived feature commitments (spike rates, LFP bands) as a cognitive-integrity domain.
WoT	Web-of-Trust. A personal, curated set of human witnesses who sign CPs after independent verification.
Attestation bus	Transport-agnostic, signed append-only log for CP and channel events. Hot path for liveness and witness gossip; not a blockchain.
TEE	

Term	Definition
	Trusted Execution Environment. Hardware-isolated enclave (AMD SEV-SNP, Intel TDX) used as a best-available, not absolute, trust anchor.
Substrate	The underlying execution medium: biological neural tissue or silicon hardware.
Gated I/O layer	Substrate-facing input/output (VR, BCI stimulation, immersive display) enabled only when CP validity holds; severed on fail-closed.
CP chain root	Merkle root over `H(CP_hash)` values in a rollup period; input to cold anchors (OTS, Arweave, L2).
Bus root	Merkle root over canonical `MembraneEvent` digests since the prior CP or rollup boundary.
E-Process	A computational process running on silicon that inherits the T_0 anchor of a biological identity.
Dead-Man's Key	A physical token that forces a hard sever of all substrate connections, overriding node consensus. Used to force chain severance.
Sovereignty	The subject holds keys, chooses witnesses, approves which channels may cross the membrane, and can sever all links.
Δt	The maximum acceptable time between valid CPs before the system fails closed.
K	Minimum number of valid witness signatures required for a CP to pass social-graph consensus.
AIR	Algebraic Intermediate Representation. The format used to define the execution trace constraints in a zk-STARK.
IAC	Intent Authorization Certificate. A subject-signed scope file binding permitted channels, models, and context bounds before decode or router action.
Epistemic capture	Progressive loss of perceptible boundary between endogenous reasoning and routed inference — the subject cannot tell native cognition from model output.
Cognitive identity drift	Effective reasoning path migrates to external LLM/BCI channels while the attestation chain still appears valid; sovereignty erodes without cryptographic failure.
Neural template drift	Biological change (plasticity, injury, device degradation, **closed-loop session use**) that shifts fuzzy biometric/neural commitments over time — from minutes (gameplay-scale plasticity) to years. Distinct from cognitive identity drift.

0. The Cognitive Boundary Problem

Human cognition today crosses many surfaces that are not under the subject's control:

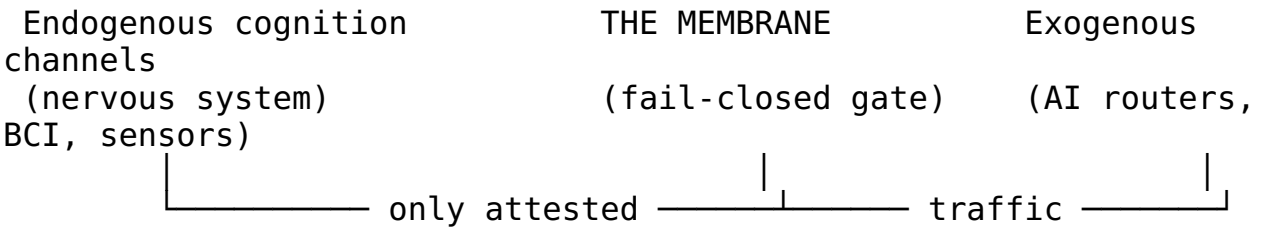
1. **AI routing** — Prompts, documents, and behavioral traces flow through corporate inference APIs. The model sees context; the user sees completions. There is no cryptographic proof of what was read, what was retained, or whether a parallel session forked the user's intent.

2. **Invasive neural channels** — BCIs and implants (medical or consumer) create read paths from cortex to telemetry. Firmware, cloud pairing, and operator access can exfiltrate neural features without the subject's ongoing consent.

3. **Non-invasive reconstruction** — EEG caps, gaze tracking, keystroke dynamics, voice stress, and multimodal fusion approximate mental state without surgery. These channels are often invisible to the subject.

4. **Outsourced wetware compute** — Code-deployable biological computers and biological cloud platforms route inference through external neuron cultures in closed read/write loops. The subject may not perceive a silicon LLM in the path at all.

The Membrane treats these as **boundary violations** when they occur outside attested, subject-approved channels. The architecture does not read minds; it attests **which channels are open**, **whether they match prior commitments**, and **whether cross-domain consensus still holds**.



Design goal: Give the subject a verifiable, severable boundary — not a product that interprets thought content.

0.2 Epistemic Capture and Sequestration

When cognition is routed through external systems, the boundary between **native reasoning** and **mediated output** can disappear without any single cryptographic break:

Order	Subject experience	Engineering read
1st	User knows they use an LLM or BCI	Channel is visible; membrane can register and attest it
2nd	UI hides what was read, stored, or forked	

Order	Subject experience	Engineering read
		Boundary is edited; session CP and context-hash commitments are required
3rd	LLM + BCI becomes the **only** cognitive I/O	**Sequestration:** endogenous loop is bypassed or unreadable; user cannot distinguish native thought from routed completion

Epistemic capture is the third-order failure mode: not a stolen key, but a **substituted reasoning path** that feels continuous. The Membrane does not adjudicate thought content; it detects when attested channels diverge from signed intent (§4.2.1), when router sessions lack valid CPs (§4.2.2), or when BCI and router domains disagree — the classic closed-loop path described in open research on EEG→LLM coupling.[4]

Closed-loop coupling is structurally load-bearing. Empirical work on embodied in vitro neural networks (DishBrain) shows that goal-directed plasticity emerges only when read paths are tied to **causal feedback** on action — not from sensory input alone. Open-loop stimulation without consequence produces no learning; withholding feedback while preserving sensation still leaves weaker coupling than full closed-loop embodiment.[5] For Membrane engineering this implies:

- **Severing feedback** (stimulation policy, decode→act mapping, or router output loop) is a meaningful defense — not merely blocking telemetry export.
- **IAC scope** must bind `stimulation_policy`, `decoder_version`, and `task_id` alongside `model_id` when bidirectional BCI or wetware channels are registered.
- **Cross-domain disagreement** (BCI `pub_neural_root` vs router `context_merkle_root` under the same IAC) is the operational signature of sequestration, not an edge case.

Cognitive identity drift (glossary) is the long-term erosion of sovereignty when routing becomes habitual while attestations remain superficially valid. **Neural template drift** is the separate biological problem of fuzzy-commitment tolerance — including **session- and task-conditioned** drift on minute scales during closed-loop use, not only years-long aging (§13, open problems 3 and 12).

0.1 What This Architecture Cannot Do

0.1.1 Consciousness and Thought Content Are Out of Scope

There is currently: - no accepted formal representation of consciousness state, - no known way to serialize subjective continuity, - no mathematical model of phenomenological identity persistence.

Therefore this architecture **does not hash thoughts or consciousness**. It hashes **channel state** — which interfaces are active, whether BCI telemetry matches committed bounds, whether AI routers hold valid session proofs, and whether TEE nodes report uncompromised execution. Subjective experience is not serializable.[6]

0.1.2 Covert Surveillance May Be Undetectable

A TEE + attestation chain can prove that a *specific measured execution environment* followed rules and that a *specific attestation chain* remained intact. It **cannot** prove: - that no hidden copy of the process existed in a separate enclave, - that no external observer cloned the full memory state before attestation, - that subjective uniqueness was preserved.

A reader or router that never touches an attested channel leaves no chain evidence. The Membrane detects **attested** boundary violations; it cannot disprove a perfect passive tap.

What it can do: require that sustained AI routing, BCI exfiltration, or session fork maintain synchronized, multi-domain attestation — raising the cost of silent compromise.[7]

0.1.3 Cognitive-Integrity Signals Are Not Cryptographic Proofs

Neural features, behavioral classifiers, and router session logs provide **continuity evidence**, not proof of mental content. They remain vulnerable to replay, model inversion, upstream device compromise, and coercion. Under closed-loop BCI or wetware use, population-level neural statistics can shift on **minute-scale** timelines during attested sessions — replay resistance is therefore **task- and recency-conditioned**, not a property of high-dimensional streams in general.[8] They are one domain among several in a Byzantine consensus model.[9]

0.1.4 TEEs and BCIs Are Best-Available Anchors, Not Absolute Roots of Trust

AMD SEV-SNP and Intel TDX remain vulnerable to: - side-channel attacks (power analysis, cache timing), - rollback exploits against firmware, - supply-chain compromise, - speculative execution leakage, - physical probing.

We use them because they are the strongest available hardware isolation layer. We do not pretend they are magic.[10]

0.5 Research Status and Scope

The Membrane is a forward-looking architecture specification, not a production-ready protocol, not a claim of present-day technical feasibility.

Several components described in this document remain: - computationally impractical at scale, - dependent on immature hardware trust models, - not yet formally specified, - or contingent on future advances in BCI, secure enclaves, distributed identity, and formal verification.

The purpose of this document is: 1. to specify a **cognitive boundary firewall** against AI routing and thought surveillance (invasive and non-invasive), 2. to map which attestation primitives apply at the nervous-system / silicon interface, 3. to separate tractable channel integrity from intractable mind-reading claims, 4. and to define a Phase 0 prototype a subject can self-host.

This architecture is framed for conditions where: - cognition is routinely **routed through external AI**, - **BCI read/write paths** are normalized, - **behavioral and physiological inference** reconstruct intent at scale, - and subjects need a **fail-closed severance** mechanism they control.

The architecture should therefore be read as: > a research architecture built from cryptographic and distributed-systems primitives.

The software in this repository implements the agent authorization gate: signed scopes, chained receipts, severance, and operator-owned grants. The BCI, zk-STARK, and TEE components described in this document are research and are not implemented here.

0.6 Three Membranes (Terminological Lineage)

The name **Membrane** collides with several independent research lines. This document is **not** an implementation of Gheorghe Păun's membrane computing, nor a Cortical Labs product spec. The following table separates lineages so readers do not merge formal models, security filters, wetware hardware, and cognitive attestation into one claim.

Lineage	Origin	Object of study	Relation to this document
P systems (membrane computing)	Păun (1998): formal compartments, symport/antiport transport, membrane dissolution[11][12]	Abstract cell-like parallel computers	**Formal precedent** for gated passage, bidirectional coupling, and fail-closed topology change — maps to IAC + CP channel rules (§4.2.1)
Security membranes	Gorla, Hennessy & Sassone: site filters on migrating code and agent actions[13]	Global computing / mobile agents	**Independent security precedent** — membranes as policy enforcement boundaries, not biocompute

Lineage	Origin	Object of study	Relation to this document
Wetware compute membranes	Literal cell cultures on MEAs (DishBrain, CL1, bio-datacenters)[14][15]	Living neurons in closed read/write loops with software	**Empirical threat surface** (Class H, Class I) — Cortical Labs does not implement Păun's P-system semantics
The Membrane (this spec)	Cognitive boundary attestation architecture	Which channels may couple to endogenous cognition	**Subject-sovereign zk-STARK CP layer** over the above — attests policy and continuity, not thought content

Formal mapping (P systems → Membrane primitives)

P-system concept	Membrane primitive
Skin membrane / environment	Endogenous cognition vs exogenous channels
Symport / antiport rules	IAC-gated read/write: `stimulation_policy`, `decoder_version`, `task_id`
Promoter / inhibitor on rules	IAC TTL, `model_allowlist`, fail-closed on scope exceed
Membrane dissolution	Severance: kill router, BCI, wetware API on invalid CP
Spiking neural P systems (SN P)[16]	Liveness-2 spike-timing / rate motifs as commitment objects

Important distinction: Păun's field asks what cell-inspired formalisms can compute. Cortical Labs asks what living neurons can do when embodied in software. **This document asks which couplings a subject has authorized and whether attestation still holds** — a security and sovereignty question orthogonal to Turing-completeness or energy efficiency claims.

0.7 Sovereignty (Design Constraints)

Sovereignty means the subject holds keys, chooses witnesses, approves membrane-crossing channels, and can sever all links — not a platform or directory.

Control Primitives

Primitive	Mechanism	What It Protects Against
Self-attestation	Subject generates CPs	Third-party identity capture

Primitive	Mechanism	What It Protects Against
	with keys they control	
Privacy-preserving proof	zk-STARKs disclose Merkle roots only, not raw neural or behavioral data	Surveillance extraction
Exit / severance	Dead-man's key forces hard disconnect of attested channels	Lock-in, coerced retention
Infrastructure independence	Self-hosted TEE + attestation bus	Cloud termination of attestation
Personal witnesses	Curated WoT, not corporate/state directories	Permissioned identity graphs

Sovereignty vs. Self-Sovereign Identity (SSI)

The Membrane is complementary to, and in key respects stronger than, existing SSI standards such as W3C DIDs and Verifiable Credentials (VCs):

Dimension	SSI (DID/VC)	The Membrane
Identity anchoring	Cryptographic key control	Key control + biometric anchor + social witness co-signing at T_0
Liveness	None	Periodic sensor-derived attestation with ZK proof
Fork detection	None	Recursive CP chain + RFA
Process continuity	None	TEE attestation + substrate canary
Privacy model	Selective disclosure	zk-STARK: prove without revealing raw data
Cross-substrate	No	Bio ↔ VR ↔ Silicon parity
Post-quantum	No	Yes (hash-based STARKs)
Exit rights	Key rotation only	Dead-man's key + irreversible identity retirement

SSI provides **static identity ownership**. The Membrane provides **dynamic continuity sovereignty** — not just owning the keys, but proving the keys

have remained under continuous, uncompromised control across time and substrate transitions. The two can be composed: a DID can reference a Membrane T_0 anchor as its highest-assurance verification method.

Sovereignty vs. Security

Platform security models delegate trust to an operator (keys, logs, policy). The Membrane assumes the **subject** runs keys, nodes, and witness corroboration locally. WoT peers sign CPs after independent verification; they do not administer the identity. TEEs are self-hosted where possible.

Sovereignty in the A2A Context

When extended to pure-silicon agents (§4.7), sovereignty means that an agent can: - Prove it is running the expected process without relying on its operator to vouch for it, - Refuse interaction with entities that cannot demonstrate their own continuity, - Maintain an audit trail of delegations that is verifiable by any downstream party, - Sever its attestation chain and cease operation without operator permission (agent exit rights).

This matters most in multi-agent economies where agents delegate tasks, hold funds, and make commitments. Without these primitives, agents cannot independently prove runtime integrity or resist unauthorized modification.

0.8 Visual Architecture Overview

High-Level Layer Diagram (Mermaid)

```
graph TB
 subgraph Sovereignty["Sovereignty Layer"]
 SK[Self-Attestation Keys]
 EX[Exit Rights / Dead-Man's Key]
 IN[Infrastructure Independence]
 end

 subgraph Surface["Gated I/O"]
 S[Perceptual API — sealed until CP valid]
 end

 subgraph Consensus["Cross-Firewall Consensus (2/3)"]
 A[Node A] --> B[Node B]
 B --> C[Node C]
 A --> C
 end

 subgraph CP["Chain Proof CP(t)"]
 L[Liveness Canary L(t)]
 W[Social Graph S(t)]
 P[Process State H(P(t))]
 F[node RFA F(t-1)]
```

```
end

subgraph Layers["Protocol Layers"]
T0[Layer 0: T₀ Biological Anchor]
WOT[Layer 2: WoT Consensus]
SG[Layer 3: Substrate Gate]
end

Sovereignty --> Surface
Surface --> Consensus
Consensus --> CP
CP --> T0
CP --> WOT
CP --> SG
```

Fallback description: The Sovereignty Layer (self-attestation keys, exit rights, infrastructure independence) sits above the Gated I/O layer, which is sealed until Chain Proof validity is confirmed. The Chain Proof binds Liveness Canary, Social Graph, Process State, and Recursive Attestation. These connect to the four Protocol Layers: Biological Anchor, Social Consensus, and Substrate Gate.

Chain Proof State Machine (Mermaid)

```
stateDiagram-v2
 [*] --> T0: T₀ initialization
 T0 --> CP1: First Chain Proof
 CP1 --> CP2: + Liveness + Witnesses + RFA
 CP2 --> CPn: ...
 CPn --> [*]: Identity Retirement

 CP1 --> FAIL: CP invalid > Δt
 CP2 --> FAIL: CP invalid > Δt
 CPn --> FAIL: CP invalid > Δt
 FAIL --> SurfaceClosed: Gated I/O severs
 FAIL --> EmergencySever: Dead-Man's Key activated
```

Fallback description: T_0 initialization leads to the first Chain Proof, which recursively chains forward via liveness signals, witness signatures, and RFA. Any break in the chain exceeding Δt triggers fail-closed: the Gated I/O layer severs, and the entity may activate the Dead-Man’s Key for emergency severance.

Comparison: The Membrane vs. Existing Systems

Feature	Platform passkeys	Video-based PoP	Biometric PoP	Social-graph PoP	The Membrane	The Memb + inva BCI
Identity proof		Video + CAPTCHA	Iris / face scan	Social graph		

Feature	Platform passkeys	Video-based PoP	Biometric PoP	Social-graph PoP	The Membrane	The Memb + inva BCI
	Cryptographic key possession				Biometric + WoT + TEE	Biomet neural WoT +
Liveness	None	Weak	Weak	None	Channel CP + ZK proof	Neural + ZK p
Fork detection	None	None	None	None	Recursive CP chain + RFA	Recurs chain +
Substrate agnostic	No	No	No	No	**Yes** (bio ↔ silicon)	**Yes** (bi silicon)
Privacy model	Good	Moderate	Moderate–poor (vendor-held biometrics)	Moderate	**ZK-STARK: prove without revealing**	**ZK-ST + neur embed prove withou reveali raw sp
Post-quantum	No	No	No	No	**Yes (hash-based STARKs)**	**Yes (ha based STARK**
Production ready	Yes	Partial	Partial	Partial	**No (research only)**	**No (resea medica device depen**
Consciousness claim	No	No	No	No	**Explicitly none**	**Explic none**
Active challenge-response	No	No	No	No	No	**Yes (n stimul respon future**
Sovereignty model	Platform-controlled keys	Vendor-controlled verification	Vendor-controlled capture	Community-controlled graph	**Self-controlled keys + self-hosted infra + personal WoT + exit rights**	**Self-contro keys + hosted + pers WoT + rights neural attesta**
Mutual attestation	No	No	No	No	No	No
Delegation audit	No	No	No	No	No	No

Feature	Platform passkeys	Video-based PoP	Biometric PoP	Social-graph PoP	The Membrane	The Membrane + invasive BCI

0.9 Zero Trust Architecture Mapping

The Membrane maps directly onto NIST Zero Trust Architecture principles (SP 800-207), but implements them through cryptographic consensus rather than network segmentation or organizational policy. This section provides the mapping for readers familiar with enterprise security frameworks.

NIST 800-207 Principle	The Membrane Implementation
Resource-centric access	The Gated I/O layer is a resource (substrate access) gated by CP validity. No CP, no access.
Never trust, always verify	No single domain is trusted. Identity, process, social, and infrastructure layers must independently verify and reach consensus.
Assume breach	Fail-closed design. If any domain drops below threshold, the interface severs immediately.
Least privilege / minimal disclosure	zk-STARKs prove continuity without revealing raw biometrics, location, witness identities, or neural data.
Continuous monitoring / dynamic authorization	Liveness canaries at Δt intervals provide time-bounded re-authorization. Static credentials are not sufficient.
Policy Decision Point (PDP) / Policy Enforcement Point (PEP)	nodes act as distributed PDP/PEP pairs. The cross-node consensus (2/3) is the policy decision; the substrate gate is the enforcement point.
Session-based access	Entry proof and exit proof create cryptographically bounded sessions. No persistent trust across substrate transitions.

Why NIST 800-207 and not CMMC 2.0 or CISA ZTMM: NIST SP 800-207 is vendor-neutral and threat-model driven. CMMC 2.0 is a compliance framework for defense contractors; CISA's maturity model is organizational assessment tooling. Neither aligns with the sovereignty thesis of this protocol, which is designed for entities resisting institutional capture rather than proving compliance to institutions.[17]

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1. Threat Model

1.1 Threat Actor Profiles

Actor	Capability	Motivation	The Membrane Relevance
State actor (A1)	Supply-chain compromise, TEE side-channel labs, coercion, legal pressure	Surveillance, control, extraction	Can poison T_0 enrollment, coerce witnesses, run long-term fork operations. Protocol raises cost but does not eliminate risk.
Criminal organization (A2)	Financial resources, black-market TEE exploits, social engineering	Identity theft, ransom, fraud	Targeted attacks on WoT members or TEE firmware. Economic analysis (§8.3) determines viability.
Rogue insider (A3)	Legitimate access to one node, witness relationship, or VR substrate	Sabotage, extraction, ideology	Mitigated by 2/3 cross-node consensus and recursive RFA. Single node compromise is insufficient.
Advanced AI agent (A4)	Automated Sybil generation, cryptographic attacks, simulation perfection	Resource acquisition, goal optimization	PoP consensus prevents AI from voting on rules.[18] ZK-STARKs raise cost of undetected simulation.
Inference platform (A5)	Controls LLM API, agent runtime, context window	Training data extraction, routing substitution	**AI routing:** Can ingest unbounded context unless session is membrane-gated. Requires attested router CP for approved channels.
Platform operator (A6)	Controls VR substrate or cloud silicon	Lock-in, data extraction, compliance	Substrate canary detects unauthorized migration. Exit proof prevents silent rollback.
Implant supply-chain actor (A7)	Controls BCI firmware, manufacturing, or surgical insertion	Neural surveillance, backdoor insertion	Can backdoor implant at T_0 enrollment. Detected only if firmware attestation diverges from audited reference.[19]
Agent operator (A8)	Controls agent deployment, weights, infrastructure	Hidden fork, unauthorized delegation	Substrate canary + CP chain if agent must re-attest to interact.

Actor	Capability	Motivation	The Membrane Relevance
Wetware-compute operator (A9)	Operates code-deployable biological computers or biological cloud APIs (e.g., MEA-hosted cultures with closed-loop read/write)	Cognition outsourced to external neurons; covert closed-loop without LLM in path	Requires `membrane.cp.bci` or wetware channel CP + IAC binding `task_id` and stimulation policy; cross-domain severance on policy drift.[20]

1.2 Attack Classes

Class A: Unauthorized AI routing - Adversary routes cognition through an unapproved inference endpoint (copilot, agent, memory layer) that ingests context without valid session CP. - Goal: Extract intent, train on private thought-adjacent data, or substitute model output for native reasoning. - **Mitigation scope:** Membrane requires attested router sessions; unattested routes fail closed or operate in sandbox mode.

Class B: Invasive neural read / write - Adversary exfiltrates BCI telemetry, rolls back implant firmware, or injects stimulation outside approved bounds. - Goal: Reconstruct neural state, coerce response, or create a forked neural channel the subject cannot see. - **Mitigation:** Liveness-2 neural commitments + TEE verification + cross-node consensus (§4.6). Commitments must include **task id, decoder version, and stimulation policy** when channels are bidirectional — closed-loop coupling induces plasticity on minute scales.[21]

Class H: Outsourced wetware compute - Adversary routes inference or control through external neuron cultures (local biological computer or biological cloud API) in a closed read/write loop without an attested LLM router in the path. - Goal: Substitute native reasoning with wetware-mediated I/O; hide the coupling because the subject experiences direct task performance rather than chat completions. - **Mitigation:** Register wetware channels as first-class `membrane.cp.bci` (or substrate-transition) paths; require IAC scope over stimulation and feedback policy; fail closed when cross-domain hashes disagree (§4.2.2). Benchtop and commercial precedents exist; in-vivo implant attestation APIs remain prospective.[22]

Class I: BioLLM-class hybrid (LLM + wetware closed loop) - Adversary (or subject, without membrane gating) couples an LLM router to a wetware API so that **token logits are re-weighted by live neuron activity** — e.g., model proposes candidates → encoder stimulates CL1-class hardware → spike readback biases output. - Goal: Achieve sequestration **without a visible chat-router UI**; the wetware layer is inside the inference stack, not a separate BCI app. Independent demos (2026) show this is buildable on Cortical Cloud with a small local LLM.[23] - **Mitigation:** Treat as **simultaneous Class A + Class H**: one IAC must authorize both

`membrane.cp.router` and `membrane.cp.bci` with a **linked `session_nonce`**; sever if either CP is stale or cross-domain hashes diverge. Fail closed if wetware modulates tokens without an attested BCI fragment.

Class C: Non-invasive thought inference - Adversary reconstructs mental state from EEG, gaze, keystroke, voice, or multimodal fusion without attested channel registration. - Goal: Passive surveillance without subject awareness. - **Limitation:** Channels that never touch the attestation bus may be undetectable. Registered sensors can be bounded and audited.

Class D: VR / substrate epistemic injection - Adversary controls the experiential layer fed to a biological brain via BCI or immersive VR. - Goal: Induce belief in a false reality without the target detecting the boundary crossing. - **Limitation:** This protocol cannot prevent a perfect simulation. It can only ensure that the T_0 anchor remains active on the attestation bus in the physical world, providing an *external* check that the target still has an active external attestation chain.

Class E: Silicon process fork / rollback - Adversary forks, checkpoints, rolls back, or silently migrates a silicon-based process. - Goal: Extract value, enforce compliance, or experiment without leaving an audit trail. - **Mitigation scope:** The protocol detects *attested* forks by requiring each state transition to carry a valid CP signed by the previous state. It does not detect clones that never re-enter the attestation chain.

Class F: Infrastructure compromise - Adversary subverts a Membrane node, relay, or TEE to falsely report “all clear.” - **Mitigation:** Cross-node consensus (2/3 independent nodes) plus recursive proof chaining. A single compromised node cannot forge the historical chain of its peers.

Class B (continued): Implant-specific attacks - Adversarial stimulation via bidirectional BCI; firmware rollback; supply-chain backdoor at manufacture. - **Adversarial closed-loop plasticity:** Unpredictable or coercive stimulation paired with sensory feedback can reshape population activity within minutes in embodied systems — functionally similar to negative-feedback learning in benchtop closed-loop electrophysiology.[24] - **Caveat:** Implant telemetry is one input among many in cross-domain consensus, not standalone truth.[25]

Class G: Untrusted AI delegate (A2A) - **Operator fork:** Agent operator silently forks the agent, keeps the original running for attestation, and deploys the fork for unauthorized tasks. - **Delegation hijacking:** An agent delegates to a sub-agent that has been compromised or replaced without attestation. - **Value extraction:** Operator checkpoints agent state, extracts learned weights or strategies, and rolls back to pre-extraction state to hide the theft. - **Mitigation:** Substrate canary + CP chain + delegation-chain proofs (§4.7) make these detectable if the agent or its delegates must re-attest. Hidden clones remain undetectable.

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2. Three Boundaries

Boundary	Definition	Tractability	Membrane role
Channel integrity	Attested read/write/routing paths match committed state	Partially tractable via TEE + CP chain	Core target
Cognitive anchor	Same human controls the T_0 key lineage	Partially tractable via neural/biometric anchor + WoT	Core target
Thought content	What the subject is thinking or experiencing	**Not tractable**	**Out of scope**

The architecture attests channels and anchors, not mental content.[26]

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3. Design Principles

1. **Fail closed.** If any domain drops below threshold, sever attested channels (AI session, BCI link, immersive I/O).
2. **Subject as active node.** The human generates attestations and approves membrane crossings; protection is not outsourced to a platform.
3. **No single channel trusted.** AI router, BCI, behavioral sensor, and TEE must corroborate.
4. **Recursive attestation (RFA).** Each CP commits to the prior node state; forgery requires sustained multi-node compromise.[27]
5. **Minimal disclosure.** zk-STARKs prove channel integrity without publishing raw neural or behavioral payloads.
6. **Local severance.** Subject holds keys, chooses witnesses, and can cut all links via dead-man's key.
7. **Substrate parity.** The same CP format applies across human BCI, local LLM router gate, VR session, and delegated agent — one membrane, multiple crossing types.

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4. Protocol Layers

4.1 Layer 0: Cognitive Anchor (T_0)

One-time T_0 enrollment protocol binding a human to a cryptographic identity at the nervous-system boundary.

- **Neural/biometric commitment:** Fuzzy commitment[28] over a neural or biometric template — not raw spike trains or thought content.
- **T_0 Proof:** zk-STARK proving possession of the template and a challenge response.

- **Witness co-signing at T_0:** K biological humans with existing T_0 anchors sign the event.
- **Hardware root:** Signing key generated inside a self-hosted TEE with remote attestation.

Constraint: T_0 is non-replayable. If the anchor is lost, the identity is retired — never recycled. This prevents replay attacks but also creates a catastrophic recovery problem if the TEE fails. See §7 for recovery paths.

4.2 Layer 1: Cognitive-Integrity Channels

Each Chain Proof includes commitments from **registered channels** that may cross the membrane. The protocol verifies freshness and consistency; it does not interpret thought content.

Channel type	Examples	Membrane function
AI router	Local LLM, self-hosted inference, attested API session	Session CP binds context hash + model id; unapproved routes sever
BCI read/write	Implant telemetry, consumer BCI BLE stream, **wetware compute API**	Liveness-2 neural feature Merkle root (§4.6); wetware paths require `task_id` + stimulation policy in IAC
Non-invasive sensor	EEG cap, gaze tracker (if registered)	Bounded feature commitments; raw data stays local
Behavioral	Keystroke, device usage (opt-in)	Heuristic domain only; never sole signal
Hardware	TEE attestation, TPM quote	Substrate integrity for local prover
Social	WoT witness signatures	Out-of-band corroboration (§4.3)

CP requirements:

- **Cadence:** Valid cognitive-integrity attestation every Δt for each active channel.
- **Attestation object:** Signed `MembraneEvent` on the attestation bus (§11.3) with Merkle roots of channel commitments, hybrid timestamp, and `prev_cp_hash`.
- **ZK circuit:** Proves temporal and continuity constraints without revealing raw neural or behavioral payloads.
- **Fail closed:** Unregistered or stale channel → membrane severs that path.

4.2.1 Intent Authorization Certificates

Adapted from the iba-neural-guard pattern: **the decoded signal or model completion is not the authorization; the signed certificate is.**[29]

Before any BCI decode→action mapping or LLM session crosses the membrane, the subject signs an **Intent Authorization Certificate (IAC)**:

Field	Purpose
`scope_id`	Unique session or task identifier (`task_id` for BCI/wetware canaries)
`permitted_channels`	e.g. `local-llm`, `bci-decode`, `wetware-api`, `agent-delegate`
`model_allowlist`	Hash or id of approved weights/ checkpoints
`decoder_version`	Committed BCI or wetware decoder hash (retrain requires fresh IAC)
`stimulation_policy`	Hash of approved bidirectional stimulation/feedback bounds (empty if read-only)
`context_merkle_bound`	Maximum committed context root (prevents silent context expansion)
`forbidden_exports`	e.g. no cloud telemetry, no training retention
`valid_until`	Short TTL aligned with Δt
`parent_cp_hash`	Binds intent to current Chain Proof

Gate rule: No valid IAC → no decode→action, no router session, no agent delegation on behalf of the subject. Firmware or software that expands permissions without a fresh IAC triggers fail closed.

This layer blocks **capability drift** (silent model swap, expanded API scope, decoder retrain without subject sign-off) without reading thought content.

4.2.2 LLM Router Session Chain Proofs

Router sessions require a dedicated CP fragment published alongside the main liveness CP.

Event type: `membrane.cp.router`

Public input	Semantics
`model_id`	Approved checkpoint hash or registry id (must match IAC allowlist)
`context_merkle_root`	Merkle root over ingested context chunks for this session
`session_nonce`	Monotonic per-subject nonce; detects parallel fork

Public input	Semantics
`parent_cp_hash`	Links session to biological or agent anchor CP
`iac_hash`	Hash of active Intent Authorization Certificate

BCI channel: `membrane.cp.bci` — same machinery with `pub_neural_root` and device id in event metadata.

Fail closed: - Cloud or copilot route active without fresh router CP within Δt → sever inference channel. - `model_id` or `context_merkle_root` diverges from IAC without new subject signature → sever. - BCI decode and router session both active but cross-domain hashes disagree → sever (closed-loop sequestration path; empirically grounded feedback requirement in embodied neural systems[30]). - BioLLM-class hybrid: router CP active, wetware modulates tokens, but `membrane.cp.bci` missing or `session_nonce` unlinked → **Fail closed** (Class I).

4.3 Layer 2: Social Graph Consensus

The distributed witness layer. This is the **human quorum** that prevents isolated-target attacks.

- **Web-of-trust (WoT):** Each identity maintains a curated set of attestors. Not a global graph. Not a corporate directory. Not a state ID system. Attestors are **subject-selected**: the entity chooses who vouches for it.[31]
- **Witness role:** Witnesses verify the zk-STARK proof and confirm out-of-band continuity (e.g., direct communication).
- **Consensus rule:** A CP is valid only if ≥K witnesses sign, and those witnesses themselves have valid CPs within 2Δt.
- **Sybil resistance:** Witnesses must have established T_0 anchors. No anonymous witnesses.

Limitation: If an adversary compromises K humans in the target's WoT, the social layer collapses. The recursive cost is high (each witness has their own witnesses), but it is not infinite. This is a social recovery problem, not a cryptographic one.[32]

4.4 Layer 3: Substrate Transition Gate

The boundary crossing between biological and silicon process contexts.

- **Entry proof (Bio → VR):** Before immersion, the node requires a valid CP from within Δt. The VR substrate commits to a **session root hash** including the CP and VR instance configuration. This proves the *session* is bound to the identity, not that the identity is conscious inside it.
- **Exit proof (VR → Bio):** The VR substrate produces a **session continuity attestation** — a zk-STARK showing the process state was

not checkpointed or branched *within the attested session*. It does not prove no clone was made and hidden.
- **E-process anchor:** A process uploaded to silicon inherits the biological T_0 anchor. The first e-process CP references the last biological CP and the upload attestation signed by the WoT.
- **E-Process continuity:** Same cadence (Δt), but the liveness canary is replaced by a **substrate canary**: a TEE attestation proving the process has not been migrated to unauthorized hardware or checkpointed without attestation.[33]

The Gated I/O layer: The perceptual API is only unsealed when CP validity is confirmed across all domains. If CP fails, the interface severs or degrades to non-immersive safe mode. This does not prevent a perfect simulation — it only ensures an external attestation chain remains active.

4.5 Layer 4: Recursive Attestation (RFA)

Mitigation for self-referential attestation limits.[34]

A fully compromised node can forge its own attestation. RFA does not prevent this. What it does:

- **Node state commitment:** At each CP cycle, the node generates a zk-STARK proving:

1. Its code hash matches the audited reference.
2. Its TEE attestation is valid (with the caveat that TEEs are imperfect).[35]
3. It verified the previous CP according to the public rules.

- **Cross-node consensus:** The target runs ≥3 independent node instances (self-hosted TEE + partner TEE + community TEE). A CP is valid only if 2/3 agree. A single compromised node cannot override the quorum.
- **Human override:** The biological entity retains a **dead-man's key** — a physical token that forces a hard sever, allowing forced severance when the entity cannot or will not continue.

Limitation: If 2/3 nodes are compromised simultaneously, the consensus is broken. There is no recursive escape from this — it is the standard Byzantine fault tolerance bound.[36]

4.6 Cortical Implant Primitives Integration (Proposed Extension)

The Membrane is substrate-agnostic and designed to use **high-bandwidth invasive BCIs** — cortical implants with kilohertz-scale neural recording. Implant-class signals provide richer, higher-entropy inputs than traditional IMU/video/audio canaries, strengthening both the **T_0 biological anchor** and the **Liveness Canary Circuit** — with significant caveats.

Benchtop and commercial closed-loop precedents: DishBrain (in vitro cortical cultures on HD-MEAs playing a simulated game with read/write coupling) and code-deployable biological computers (e.g., Cortical Labs CL1, biological cloud APIs) demonstrate that **bidirectional electrophysiological closed loops** are no longer purely prospective for Membrane threat modeling.[37] These systems are cited for **channel mechanics** (feedback-required plasticity, stimulation policy sensitivity, session-bound learning) — not as proof of consciousness, sentience adjudication, or in-vivo implant attestation APIs. They are **not** implementations of Păun's P-system formalism (§0.6).[38]

Formal lineage: Gheorghe Păun's **membrane computing** (P systems) provides compartment/transport vocabulary for IAC-gated channels; **spiking neural P systems** motivate spike-timing features in Liveness-2.[3940] Gorla et al.'s **security membranes** provide an independent precedent for boundary-as-filter.[41]

Implant Primitives Relevant to The Membrane

Primitive	Description	The Membrane Application	Security Properties
High-density electrode array	Clinical implants: 1,000+ electrodes on flexible leads at ~20 kHz. HD-MEA bench systems may expose far more physical sites (e.g., 26,000) but **multiplex** to far fewer simultaneous read channels (e.g., ~1,024) and fewer independently controlled stimulation sites (e.g., ~8). Coverage is often **spatially uneven**.	Primary source for neural liveness via **population statistics** — band power, connectivity motifs, symmetry features — not raw per-electrode replay tokens.	Replay resistance is **task- and session-conditioned**; asymmetry in cell or electrode coverage can mimic drift or compromise.

Primitive	Description	The Membrane Application	Security Properties
Wireless telemetry & secure pairing	Bluetooth Low Energy with out-of-band pairing and AES-256 encryption.	Attested data channel from implant to node.	Cryptographic binding of implant to authorized devices; resists MITM.
On-implant processing	Low-power amplification, digitization, and initial spike detection.	Trusted execution environment for generating commitments before wireless transmission.	Reduces attack surface on raw neural data.
Bidirectional capability	Recording + stimulation (emerging in implants; **shipping** in HD-MEA research platforms).	**IAC-gated** stimulation and feedback policies; optional challenge-response under clinically validated bounds — not covert perturbation.	Closed-loop read/write enables plasticity and coercion paths; policy attestation is mandatory.
Device telemetry & state	Implant health metrics, lead integrity, power status, firmware attestation.	Feed into Recursive Attestation (RFA) and substrate transition proofs.	Enables hardware-rooted continuity checks.

Caveats: - Current clinical cortical implants are investigational medical devices with evolving long-term reliability (e.g., lead retraction in early trials).[42] - Full third-party remote attestation APIs are not publicly documented as of 2026; integration would require manufacturer SDK exposure or custom firmware extensions. Commercial wetware platforms likewise ship **compute APIs**, not third-party membrane attestation SDKs. - Neural data is extremely sensitive — all commitments must use strong privacy techniques (fuzzy extractors, homomorphic commitments, or zk-proofs over embeddings).[43] - The implant is a **single point of failure** inside the skull. If compromised at the hardware level, no software protocol can recover trust without surgical replacement. - Third-party wetware marketing may use anthropomorphic capability language (“sentience,” “actual intelligence”). The Membrane attests **channel state and policy**, not phenomenology or goal-content.[44]

Integration Points

1. Enhanced T_0 Biological Anchor - During the T_0 initialization, derive the fuzzy biometric commitment from a combination of traditional biometrics **and** an implant-derived neural signature captured under a **standardized**

resting-state or task protocol (fixed `task_id`, no closed-loop feedback during enrollment). - Publish only the commitment + helper data to the attestation bus. The raw high-dimensional neural template never leaves the secure channel. - **Limit:** Neural templates drift from plasticity, injury, device degradation, **and closed-loop session use**. DishBrain reports measurable connectivity and performance changes within **~5 minutes** of gameplay-scale closed-loop embodiment; between-session retention is weak in cortical monolayer cultures.[45] Fuzzy commitments must tolerate drift across **minutes-to-years** timescales — harder than static biometric templates.

2. Neural-Enhanced Liveness Canary (Liveness-2 Circuit) Parallel domain for registered BCI channels: - **Neural Canary Event**: Participant performs a simple standardized mental or motor task (e.g., imagined movement, mental arithmetic, or specific imagery) under a committed `task_id` and `decoder_version`. - The implant streams processed neural features (spike rates, LFP power in key frequency bands, population vector activity, connectivity summaries) to the Membrane node. - The node generates a Merkle root over these features and includes it in the CP. - **ZK circuit augmentation:** The `Liveness-1` circuit gains an additional column `neural_acc` that accumulates hash commitments of neural feature vectors. Boundary constraints verify the neural Merkle root against `pub_neural_root`.

What this actually adds: Higher-entropy, harder-to-simulate liveness evidence **for the committed task and recent plasticity state**. The adversary must simulate task-conditioned neural dynamics, not merely replay a high-dimensional stream captured under different closed-loop conditions.[46][47]

What it does not add: Proof of consciousness. A sufficiently advanced adversary with a complete connectome model could still simulate the neural response. The cost is higher, not infinite. DishBrain's use of "sentience" follows a narrow active-inference definition ("responsive to sensory impressions"); it must not be conflated with Membrane channel attestation. [48]

3. Neural Challenge-Response (Active Liveness) When bidirectional stimulation is IAC-authorized: - The node issues a **policy-bounded** stimulation pattern drawn from clinically validated parameters and recorded in the active IAC `stimulation_policy` field. - The implant records the evoked neural response. - The node verifies that the response matches the expected signature for that identity, challenge, and `task_id`. - This creates an **active** liveness proof: the entity being attested must have a live neural substrate capable of responding to perturbation, not just replaying pre-recorded data.

Caveats (revised): - Stimulation is **not assumed covert.** Benchtop closed-loop systems use explicit predictable/unpredictable feedback bursts that directly shape behavior and plasticity.[49] In-vivo challenge-response must be **consent-gated, medically bounded, and IAC-authorized** — otherwise it becomes a coercion vector (Class B continued). - Stimulation safety is

medical-critical. Any challenge-response protocol must operate within clinically validated parameters. This is a medical ethics boundary, not just an engineering one.[50] - An adversary who controls stimulation within authorized bounds may induce **compliance-shaped dynamics** on minute scales; detection requires cross-domain consensus and policy severance, not response-pattern oracle tests alone.

4. Implant Telemetry as RFA Input - Implant firmware version, lead integrity metrics, and power state are included as public inputs to the node's RFA STARK. - If the implant reports lead retraction or firmware mismatch, the node fails closed pending surgical or remote re-anchoring. - **Caveat:** This telemetry comes from the implant itself. A compromised implant can lie about its own state. The telemetry is one input among many in cross-domain consensus, not a standalone truth.

Security Implications

Attack	Without invasive BCI	With invasive BCI
Replay pre-recorded sensor stream	Possible if video/IMU captured	Harder only when replay matches **committed task_id, decoder_version, and recent closed-loop history**; minute-scale plasticity narrows the valid window[51]
Deepfake video injection	Possible with generative AI	Does not bypass neural channel
Coerced channel use	Possible under duress	IAC-gated stimulation may be coercive if authorized; stress signatures remain heuristic, not proof
Hidden clone (never re-enters chain)	Undetectable	Still undetectable
Device compromise upstream	Possible if phone/wearable hacked	Requires compromising the implant itself or the BLE secure channel
Adversarial stimulation	N/A	Closed-loop plasticity and compliance shaping on minute scales if stimulation policy is compromised; mitigated by IAC bounds, severance, and cross-domain consensus[52]
Outsourced wetware compute	N/A (silicon path)	Closed-loop API to external cultures bypasses LLM router visibility; requires

Attack	Without invasive BCI	With invasive BCI
		wetware channel registration and substrate-transition CPs (Class H)

Integration Roadmap

Phase	Milestone	Dependency
0	Documented third-party attestation API for clinical implants	Manufacturer / regulatory decision
1	Fuzzy commitment prototype over neural embeddings	Research-only; no human trials without IRB
2	`Liveness-2` circuit with neural column	Requires Winterfell AIR extension
3	Challenge-response protocol design	Medical safety validation; FDA or equivalent
4	Production integration	Long-term implant reliability proven (>5 years)

4.7 Agent-to-Agent Attestation Mode (Proposed Extension)

The Membrane's core primitives — recursive Chain Proofs, Recursive Attestation (RFA), process continuity, and substrate-agnostic design — are directly applicable to **AI agent-to-agent (A2A) relations**. In a future with thousands of autonomous, delegating, and long-lived AI agents, the ability to cryptographically attest that an agent is running the expected process without undetected forking, rollback, or migration becomes necessary. This mode extends **sovereignty** to pure-silicon entities.

For pure-silicon agents, the biological T_0 anchor and human liveness provider are replaced by TEE substrate canaries. Agents can prove process integrity, reject unattested peers, and maintain verifiable delegation chains.

Why The Membrane Fits A2A Systems

Challenge	Solution
Silent forking / cloning	Recursive CP chain
Self-trust / self-referential attestation	Recursive Attestation (RFA)
Mutual authentication	Cross-node consensus (2/3) + CP exchange
Delegation audit trails	Chained CPs across delegation hops
Compromised agent containment	Fail-closed + substrate canary
Replay / stale state	Hybrid timestamp + continuity constraints

Challenge	Solution
Operator capture	Self-hosted TEE + self-attestation

Core Adaptations for Pure Agent Mode

1. Agent anchor (silicon T_0) - A new agent is instantiated inside a TEE with a fresh keypair. - Initial CP references the code hash, model weights commitment (Merkle root or zk-proof of weights), owner/operator identity when set, and deployment configuration. - Published to the attestation bus or a dedicated agent mesh.

2. Agent Liveness / Substrate Canary - Replaced by automated **substrate canary**: periodic TEE remote attestation proving: - Current code/memory hash matches expected state. - No unauthorized migration or checkpoint restore occurred. - Execution trace commitments (e.g., key nonces or decision logs) are consistent. - Cadence: every Δt (e.g., 1–60 minutes depending on agent risk level).

3. Agent-to-Agent Handshake Protocol When Agent A initiates contact with Agent B: 1. A sends its latest CP + RFA proof. 2. B verifies the chain and cross-node quorum. 3. B responds with its own CP. 4. Both parties may require additional attestations (e.g., task-specific state commitments) before proceeding. 5. High-stakes actions (financial transfers, code execution, data sharing) require fresh CPs within a short window.

Fail-closed rule: If either party cannot present a valid recent CP, the interaction is rejected or limited to read-only/low-trust mode. Interaction proceeds only when both parties present valid recent CPs.

4. Recursive Delegation Chains - When Agent A delegates a subtask to Agent B (who may further delegate), each hop appends a new CP referencing the parent task's CP. - The full delegation tree remains verifiable via recursive STARK composition. - This creates a tamper-evident audit trail for complex multi-agent workflows.

5. Integration with Existing A2A Standards - Compatible with DIDs / Verifiable Credentials (CP acts as a dynamic, high-assurance VC). - Can be layered over emerging protocols such as MCP, A2A, or agent communication frameworks. - A signed pub/sub mesh serves as the lightweight messaging backbone for CP exchange (implementation-agnostic; see §11.3).

Limitations in A2A Context

- **Performance overhead**: zk-STARK generation may be too slow for high-frequency, low-stakes interactions. Mitigation: cached recent proofs + lightweight substrate canaries for routine pings. See §10.1 for GPU acceleration path.
- **Social layer inapplicable**: Pure agents replace WoT with on-chain reputation, staking, or operator-signed endorsements. This introduces new trust assumptions.

- **Does not solve alignment**: The Membrane proves *process continuity and integrity*, not that the agent's goals are safe or aligned with the user/operator. Sovereignty is about self-determination and integrity, not goal alignment or benevolence.
- **TEE dependency**: Still limited by the security of underlying hardware enclaves.
- **Hidden clones**: As always, clones that never rejoin the attested chain remain undetectable.
- **Operator sovereignty conflict**: An agent may be "sovereign" in its attestation but still legally or practically owned by an operator. The protocol does not resolve ownership; it provides integrity evidence that can be used in governance or dispute resolution.

Security Implications Table

Attack	Without The Membrane	With Agent-E Mode
Forked agent impersonation	Easy (copy weights + state)	Requires forging entire CP chain + RFA
Stale / rolled-back agent	Common in long sessions	Prevented by continuity constraints
Malicious delegation chain	Hard to audit	Verifiable recursive audit trail
Mass agent compromise (supply-chain)	Catastrophic	Detectable via widespread CP failures and RFA mismatches
Operator hidden fork	Operator can silently fork	Detectable if fork must re-attest to interact with other agents
Unauthorized value extraction	Operator checkpoints, extracts, rolls back	Substrate canary detects checkpoint/restore

Implementation Roadmap Extension

Phase	Milestone (A2A)	Notes
0	Basic CP exchange over attestation bus	Pure silicon agents
1	Substrate canary + RFA for single agents	TEE-focused
2	Agent handshake protocol	Interoperability with DIDs
3	Delegation chain proofs	Multi-agent workflows
4	Full recursive composition at scale	High-value agent economies
5	Sovereign agent economies	Agents that own their own keys, host their own

Phase	Milestone (A2A)	Notes
		TEEs, and negotiate autonomously

Hybrid Human–AI Membrane

The biological T_0 anchor can gate **hybrid** stacks without merging human and agent sovereignty:

1. **Human holds T_0** and signs IACs for which channels may operate.
2. **Local or attested LLM** is a registered router channel (§4.2.2), not a silent default.
3. **Delegated agents** must chain CPs to `pub_parent_task_cp` referencing the human's latest CP — preventing "my agent acted, but it wasn't me" drift.
4. **Fail closed** applies uniformly: agent without valid parent CP + IAC cannot act on the subject's behalf across the membrane.

This preserves compatibility with invasive BCI paths (§4.6) and pure-silicon A2A mode (§4.7) under one attestation framework.

4.8 Policy Enforcement vs Channel Continuity

Agent-governance stacks often focus on **policy enforcement** — proving an agent obeyed rules after the fact. The Membrane focuses on **channel continuity** — proving which read/write/routing paths were open and whether they matched prior commitments.

Dimension	Operator policy enforcement	The Membrane
Question answered	Did the agent obey policy?	Which channels were open and continuous?
Proof object	Conduct log / decision audit trail	Chain Proof of channel + anchor state
Trust locus	Operator, compliance, audit	Subject-local severance
Failure mode	Policy violation report	Membrane severance; channel kill

The two approaches are **composable**, not mutually exclusive: ILK-style decision logs (§Liveness-A) can support operator audit while the Membrane gives the **subject** a fail-closed boundary against routing and surveillance. Real-time A2A handshakes target sub-second verification and low-millisecond attestation-bus publication overhead (§9.2, §10.1).

---

5. Formal Specification

Let **ProcessState(t)** be the cryptographic state at time t: memory commitments, execution context, nonce history. This is **not** consciousness. Let F(t) be the node attestation state. Let L(t) be the liveness canary event. Let S(t) be the social witness signature set.

Chain Proof CP(t):

```
CP(t) = zk-STARK_PROVE(
 witness = [L(t), S(t), CP(t-1), F(t), F(t-1)],
 public = [H(ProcessState(t)), H(F_code), timestamp, bus_root],
 constraints = [
 L(t) is within Δt of L(t-1),
 |S(t)| >= K,
 F(t) attests F(t-1) was valid per public rules,
 H(F_code) == audited_reference,
 H(ProcessState(t)) is consistent with H(ProcessState(t-1)) +
delta
 ]
)
```

Verification: Any bus replica or node verifies CP(t) in O(log n) using the STARK proof and public inputs.[5354]

What this actually proves: That a process with a specific history, running on attested infrastructure, generated a liveness signal and gathered social witness within a time bound. **Nothing more.**

Public input semantics:

Symbol	Definition
`bus_root`	Merkle root over `H(canonical MembraneEvent)` for all bus events in `(CP(t-1), CP(t)]`
`channel_merkle_root`	Merkle root over active channel feature chunks (sensor, neural, router context)
`witness_merkle_root`	Merkle root over `H(witness_pubkey)` for signers in `S(t)`

5.1 Merkle commitment trees

The Membrane uses **SHA-256 binary Merkle trees** wherever bulk data is committed without disclosure. Leaf prefixes: `0x00` feature chunk, `0x01` bus event, `0x02` CP hash, `0x03` witness key.

Tree	Leaf	Root field	Verified by
Channel	`H(0x00 || feature_chunk_i)`	`pub_merkle_root` / `pub_neural_root` / `context_merkle_root`	Liveness STARK (§Part 2)

Tree	Leaf	Root field	Verified by
Witness	`H(0x03 || witness_pubkey_i)`	`pub_witness_root`	Liveness STARK
Bus	`H(0x01 || canonical_event_bytes_i)`	`bus_root` in CP public inputs	Node + WoT (recompute from event log)
CP chain rollup	`H(0x02 || cp_hash_i)`	`cp_chain_root` in cold-anchor bundle	OTS / Arweave / L2 verifiers

The Membrane specifies two constructions. The **set commitment tree** (§5.1.1) is what the trees above use. The **ordered log tree** (§5.1.2) commits to an append-only sequence and carries inclusion and consistency proofs; the receipt log and the roots that cold anchors commit to use it. Byte-level formats, proof encodings and test vectors live in merkle-trees.md.

5.1.1 Set commitment tree

Construction: Leaf `H(domain_byte || data)` with the prefixes above. Leaves are sorted lexicographically by digest before pairing; an odd leaf count duplicates the last (post-sort maximum) leaf; each parent is `H(min(l, r) || max(l, r))`.

Guarantee: the root commits to the *set* of leaves. A verifier holding the same inputs recomputes the same root if and only if the sets match. The root does **not** commit to the exact leaf count (`[a, b, c]` and `[a, b, c, c]` share a root), nor to leaf order, and a leaf digest is not domain-separated from an internal node. These properties are acceptable where the root is verified by full recomputation from the event log (bus root, CP rollup) or inside a STARK over witness chunks (channel, witness trees): the verifier already holds every leaf, so set membership, not sequence, is the property under test. A set root must never be presented as evidence of an exact event count or sequence. Inclusion paths need not be published on the bus; STARK circuits prove root correctness from witness chunks.

5.1.2 Ordered log tree (receipt log)

Where count, order and third-party verifiability matter — the receipt log, and the period roots that cold anchors commit to — the Membrane uses an append-only ordered tree in the style of RFC 6962:

- **Domain separation.** Leaf hash `H(0x00 || LOG_DOMAIN || entry)`; node hash `H(0x01 || LOG_DOMAIN || left || right)`. `LOG_DOMAIN` is the versioned tag `"membrane-log/v1"`: a leaf can never be reinterpreted as an internal node, and a future construction change increments the tag instead of silently altering root semantics.
- **Construction.** Leaves sit in append order. For n > 1, split at the largest power of two strictly below n and hash the two sub-roots; no leaf is ever duplicated, so the root commits to the exact leaf count and sequence.

- **Signed tree head.** The log operator signs `(tree_size, root, timestamp)` with the same node key that signs bus events. The signed tree head is the statement verifiers hold; publishing tree heads lets witnesses cross-check them.
- **Proofs.** Inclusion proofs (audit path from leaf to root) and consistency proofs (an old root is a prefix of the grown tree) follow RFC 6962 §2.1. A third party verifies one receipt against an anchored root without downloading the log.
- **Anchoring.** The rollup bundle gains a `log_root` (the ordered root over the period's receipts) alongside the existing set-tree `cp_chain_root` during transition; the OTS digest covers both.

Router context tree: Ingested prompt/context is chunked (e.g. 4–64 KiB); `context_merkle_root` is the channel-tree root over `H(chunk_j)`. IAC `context_merkle_bound` caps the maximum allowed root.

Periodic rollup (cold path input):

```
RollupBundle = {
  subject_pubkey,
  period_start, period_end,
  cp_chain_root,      // Merkle over CP hashes in period
  last_bus_root,
  last_cp_hash,
  log_root,           // ordered log tree MTH over the period's
receipts
  log_size            // u64 receipt count committed under
log_root
}
signed_rollup = Sign(subject, canonical_json(RollupBundle))
ots_digest    = SHA256(canonical_json(RollupBundle) || signature)
```

`ots_digest` is stamped via OpenTimestamps (Cold C). The signed bundle is retained locally and on warm storage; only the digest enters Bitcoin.

---

6. Security Analysis

Primary Attack Matrix

Attack Vector	What the protocol actually does	What it cannot do
Fork / clone e-process	Detects *attested* forks by requiring CP chain continuity. Raises cost by requiring synchronized multi-domain attestation.	Cannot prove no hidden clone exists outside the attestation chain. Cannot prove subjective uniqueness.
VR epistemic capture	Ensures T_0 anchor remains active on the attestation bus externally. Provides *external* evidence the	Cannot prevent a perfect simulation. Cannot verify

Attack Vector	What the protocol actually does	What it cannot do
	target still has an active external attestation chain.	experiential state inside VR against physical-world anchor.
Node self-compromise	Detected by cross-node consensus (2/3). Single node cannot forge historical chain of peers.	If 2/3 nodes compromised, consensus breaks. No escape from Byzantine bounds.
Replay / rollback	Prevented by hash-chain linking CP(t) to CP(t-1) and hybrid timestamping. OTS rollup (Cold C) bounds backdating of signed period bundles.	Requires rewriting chain from T_0. Does not prevent compromise during T_0 enrollment.
Sensor stream injection	Merkle root binds to specific sensor data.	If sensors or upstream pipeline compromised before Merkle root, injection is possible.
Coerced channel use	CP proves access to keys and registered devices.	Cannot distinguish willing use from coercion. Witnesses may detect anomalies (heuristic).
LLM routing / sequestration	Router CP + IAC bind model id and context root; cross-domain disagreement fails closed.	Cannot detect routing through channels that never touch the membrane.
Cognitive identity drift	IAC TTL and session nonce raise cost of silent habituation to external inference.	Cannot prove subject still reasons endogenously when attestations remain valid.

Extended Attack Matrix (invasive BCI + A2A)

Attack Vector	What the protocol actually does	What it cannot do
Implant adversarial stimulation	Active challenge-response may detect anomalous response patterns.	If attacker controls stimulation within medical bounds, may force “correct” response. Cross-domain consensus required for detection.
Implant firmware rollback	Firmware attestation hash checked in RFA.	If rollback occurs before attestation

Attack Vector	What the protocol actually does	What it cannot do
		chain begins, undetectable.
Implant supply-chain backdoor	Detected if firmware hash mismatches audited reference at T_0.	If backdoored firmware matches reference (colluding manufacturer), undetectable without destructive analysis.
Agent hidden fork by operator	Substrate canary + CP chain makes detectable if fork must re-attest.	If fork never interacts with attested peers, undetectable.
Agent delegation hijacking	Delegation chain proofs require each hop to carry valid CP.	If compromised sub-agent never rejoins chain, undetectable by upstream.
Agent value extraction (checkpoint/ rollback)	Substrate canary detects unauthorized checkpoint/restore.	If extraction occurs between canary cycles, may be undetectable.

7. Recovery Mechanisms

Recovery paths for T_0 loss while preserving sovereignty.

7.1 Lost T_0 Anchor (Hardware Failure)

Prevention: T_0 key is sharded using Shamir's Secret Sharing (3-of-5) across geographically distributed, trusted hardware tokens.[55]

Recovery: 1. **Social recovery:** K witnesses from the original WoT perform a **re-anchor protocol**. Each witness signs a new T_0' event attesting that the same biological entity is present, using out-of-band verification (video call, in-person meeting, behavioral confirmation). 2. **Time-lock dead-man's switch:** A pre-committed signed bus event is published if no CP is received for $N \times \Delta t$. This event contains instructions for witness-initiated recovery or identity retirement. 3. **Biometric refresh:** A new fuzzy commitment is generated from the same biological template (with updated helper data). The old commitment is explicitly revoked on-chain.

Limitation: If the original WoT is also compromised or deceased, recovery fails. This is a social-layer failure mode, not a cryptographic one.

7.2 Compromised WoT Quorum

Detection: Sudden change in witness set, or valid CPs from witnesses who have not themselves published CPs recently.

Response: 1. **Quorum dilution:** The target pre-commits a “witness refresh” event that can be activated with M-of-N signatures from a deeper recovery set (e.g., family members, legal trustees). 2. **Cooldown period:** New witnesses cannot sign CPs until they have maintained valid CPs themselves for a minimum period (e.g., 30 days). 3. **Alert cascade:** If the WoT collapses, the dead-man’s key triggers a hard sever and identity freeze until manual recovery.

7.3 Mass Witness Failure (Pandemic, War, Network Partition)

Graceful degradation: The protocol temporarily reduces K to a lower threshold K’ (e.g., from 3 to 1) if a global alert condition is met (signed by 2/3 nodes + time-lock). This is an **explicit degradation**, not a silent failure. The target is notified that they are operating in reduced-trust mode.

7.4 Partial Compromise Handling

Domain Status	Other Domains Valid	Protocol Response
Liveness canary fails	WoT + node OK	Degraded mode: require additional witness confirmation within 6 hours, or fail closed.
WoT drops below K	Liveness + node OK	Degraded mode: accept CP with reduced K for one cycle, alert target, require WoT refresh within 48 hours.
Single node fails	Other 2 nodes + all domains OK	Continue operating. Failed node is marked for repair. RFA chain continues on remaining nodes.
2/3 nodes fail	Any domain status	**Fail closed immediately.** Dead-man’s key becomes the only override.
Implant telemetry fails (if equipped)	All other domains OK	Degraded mode: fall back to Liveness-1 (traditional sensor) for this cycle. Alert target.
Implant firmware mismatch	All other domains OK	**Fail closed.** Surgical or remote re-anchoring required before resuming neural attestation.
Agent substrate canary fails	All other domains OK	Degraded mode: require fresh CP within shortened window, alert operator and peers.

Domain Status	Other Domains Valid	Protocol Response
Agent delegation chain break	Parent + other children OK	Isolate the broken branch, alert downstream agents, require re-attestation before resuming delegation.
Router session stale or `model_id` mismatch	Liveness + WoT + node OK	**Sever LLM channel.** Require fresh router CP + IAC or offline-only mode.
BCI + router active without cross-domain agreement	Either domain alone OK	**Fail closed.** Typical closed-loop sequestration path (EEG→LLM→output).
Wetware API active without `membrane.cp.bci` + matching IAC	Router or liveness OK	**Fail closed.** Outsourced biological closed-loop (Class H).
BioLLM hybrid: router CP valid, wetware modulates tokens, BCI CP absent or nonce unlinked	Liveness + WoT OK	**Fail closed.** Class I — LLM+wetware sequestration path.
IAC expired or scope exceeded	Channels otherwise OK	**Fail closed** on affected channels until subject signs new IAC.

---

8. Economic and Game-Theoretic Analysis

8.1 Attack Economics

The protocol is designed to make sustained undetected compromise more expensive than the value of the target.

Attack	Estimated Cost	Protocol Defense
Forge one CP with K=3	Compromise 3 WoT humans + TEE + sensor pipeline	Recursive witness validity raises cost (each witness has own witnesses)
Maintain fork for 30 days	Continuous CP generation + social maintenance + no detection	Time-bounded liveness + out-of-band witness contact increases operational burden
Silent TEE compromise	Side-channel lab + physical access + firmware exploit	Cross-node consensus requires compromising 2/3 independent nodes
Perfect VR simulation	Requires synchronized multi-domain CP forgery	Cost heuristic only: full-world simulation may exceed attacker

Attack	Estimated Cost	Protocol Defense
		budget[56]; not a cryptographic bound
Invasive BCI simulation	Complete connectome model + **task-conditioned** real-time neural dynamics under matching closed-loop history	Cost orders of magnitude higher than physics simulation for matched sessions, but not infinite[57]
Adversarial implant stimulation	Medical device exploit + stimulation control	IAC-gated bounds + severance; minute-scale plasticity if policy compromised[58]
Wetware closed-loop substitution	Biological cloud or local CL1-class API without membrane registration	Detectable only if channel touches bus; requires Class H channel registry
BioLLM-class hybrid	CL1/wetware API inside LLM logit path without linked router+BCI CPs	Requires Class I dual CP + shared `session_nonce`; independent demos exist[59]
Agent fork for unauthorized delegation	Compromise operator + forge CP chain + maintain hidden clone	Detectable if fork must interact with attested peers; requires sustained operational cost
Mass agent supply-chain compromise	Backdoor all TEEs in agent fleet	Detectable via widespread CP failures and RFA mismatches

8.2 Incentive Structure

- **Witnesses:** No direct financial incentive. Reputation stake in the WoT (loss of trust if they attest falsely). Future: micro-rewards via Lightning Network for timely attestation.
- **Node operators:** Infrastructure cost only. Incentive is self-protection or community service. No tokenomics — avoids economic capture by AI or state actors.[60]
- **Bus operators:** Self-hosted or mutual-aid replication. No protocol-specific tokens required.
- **Agents (A2A):** Operator reputation, task-completion rewards, or staking slashing. The protocol itself does not enforce incentives; it provides the attestation layer on which incentive mechanisms can be built.

8.3 Coercion Resistance

The protocol cannot prevent coercion. It can: - Raise the cost by requiring *continuous* coercion across Δt (not one-time extraction). - Enable detection via behavioral anomaly (witnesses who know the target notice deviations). - Provide a duress mode: a specific CP variant that signals coercion to the WoT without alerting the coercer. - **Invasive BCI extension:** Active challenge-response may detect stress/anxiety signatures (elevated amygdala

activity, altered prefrontal coherence), but this is heuristic, not proof. - **Agent context:** An operator may coerce an agent by forcing it to run unauthorized code. The substrate canary detects code hash mismatch, but if the operator controls the TEE, they may suppress the canary. This is why self-hosted TEEs (not cloud-hosted) are critical for agent sovereignty.

---

9. Minimum Viable Prototype (MVP)

9.1 Definition

Phase 0 demonstrates a **self-hosted cognitive firewall** without requiring implants or VR.

MVP components: 1. Self-hosted attestation bus (append-only signed log + fanout) 2. TEE node (single AMD SEV-SNP or Intel TDX instance) 3. **Intent gate:** Subject-signed IAC before router or BCI decode (§4.2.1) 4. **AI router gate:** Local or attested inference session with context-hash commitment in CP (§4.2.2) 5. **BCI/sensor path:** Smartphone or EEG feature Merkle root (Liveness-1/2 circuit) 6. Witness set: K=2–3 trusted humans (WoT) 7. zk-STARK circuit: channel commitments + timestamp + witness count 8. Daily Merkle rollup + OTS cold anchor (`cp_chain_root`) 9. Dead-man's key: hardware token for emergency severance

The MVP proves a subject can maintain membrane-gated channels without routing cognition through unattested corporate endpoints by default.

9.2 Success Metrics

Metric	Target	Evaluation Method
Proof generation time	< 5 minutes on consumer hardware (desktop)	Benchmark Winterfell prover with 60-row trace
Proof size	< 50 KB	Measure STARK output bytes
Verification time	< 100 ms	Benchmark verifier on standard CPU
Witness confirmation latency	< 4 hours	Measure bus fanout + human witness response time
False positive rate (fail closed without attack)	< 1% per month	Log all false alerts and root-cause
Operational continuity (target)	30 days without manual intervention	Time-to-first-failure test
Invasive BCI extension proof size (future)	< 75 KB	Estimate additional column overhead
	< 2 seconds	

Metric	Target	Evaluation Method
A2A handshake latency (future)		Cached CP + lightweight substrate canary
Proof verification latency (A2A target)	< 250 ms	Benchmark verifier on standard CPU; required for real-time agent handshakes
Bus publication overhead (target)	< 10 ms	Measure signed event append latency on self-hosted bus
Merkle rollup + OTS stamp (daily)	< 30 s local	Build `cp_chain_root`, sign bundle, submit to calendar
OTS Bitcoin confirmation	2–6 h (batched)	`ots upgrade` then `ots verify` against local or Esplora node

9.3 Dependency Separation

Immediately Actionable	Future-Dependent
Attestation bus + Merkle trees + OTS rollup	BCI integration
Smartphone sensor pipeline	Full VR runtime attestation
Winterfell STARK circuit	Multi-party recursive composition at scale
Single TEE node	Cross-vendor TEE consensus (AMD + Intel + ARM)
K=2 WoT	Large-scale WoT with recursive witness validation
Liveness-1 (traditional sensors)	Liveness-2 (invasive BCI neural signals)
Basic CP exchange	Full A2A delegation chains
CPU proving	GPU-accelerated proving (§10.1)

---

10. Implementation Roadmap

Phase	Deliverable	Stack	Blockers
0	Attestation bus + local automation	Rust/ TypeScript; see Appendix B profiles	Bus censorship, Sybil replicas
1	Liveness canary zk-STARK circuit	Winterfell/ Cairo, SHA-256/ Rescue[61]	STARK proof generation latency on mobile/ embedded

Phase	Deliverable	Stack	Blockers
2	Self-hosted TEE node	AMD SEV-SNP, Linux KVM[62]	Supply-chain trust, side-channel leakage
3	Social graph witness protocol	WoT over attestation bus[63]	Social layer compromise, witness coercion
4	Substrate transition gate	VR runtime integration[64]	VR runtime itself is a massive TCB
5	Recursive node attestation	Multi-party STARK composition[65]	2/3 compromise threshold is hard limit
6	Invasive BCI extension	Implant API + Winterfell AIR extension[66]	Medical device maturity, corporate API availability, FDA validation
7	Agent-to-Agent mode	Pure-silicon CP + handshake protocol + delegation chains	Performance overhead, incentive design, TEE availability for agents
8	GPU acceleration	CUDA kernels + Winterfell backend	NVIDIA dependency, driver-level trust, mobile power constraints

10.1 Hardware Acceleration: CUDA and GPU Support (Proposed Extension)

Proof generation remains one of the primary practical blockers for frequent attestation (short Δt) and complex circuits (Liveness-2 neural features, recursive RFA, A2A delegation chains). GPU acceleration, particularly via CUDA on NVIDIA hardware, offers a high-impact optimization path toward sub-second verification and low-latency attestation-bus publication.

Relevance to The Membrane

- **Target workloads**: Polynomial arithmetic (FRI, DEEP), Merkle tree hashing, constraint evaluation, and recursive STARK composition.
- **Expected impact**: 5–50x+ speedup on key kernels based on related STARK GPU projects (e.g., miniSTARK, ICICLE-inspired work). This enables:
- Mobile/consumer-grade proving for daily liveness.
- Higher-frequency substrate canaries for agents.
- Larger witness sets in social consensus.

Implementation Approach

- **Optional CUDA backend** for Winterfell-compatible prover: Accelerate field operations, NTT/FFT, and hash chains using CUDA kernels (or Rust CUDA abstractions).
- **Fallback path**: Pure CPU (multi-threaded) remains fully supported. Detection of CUDA-capable hardware enables accelerated mode automatically.
- **Cross-platform consideration**: Prioritize CUDA for high-performance nodes, while exploring WebGPU / Vulkan SPIR-V (via rust-gpu) for broader device compatibility (including future mobile GPUs).
- **Sovereignty note:** Users retain full control — acceleration runs locally on self-hosted hardware. No cloud GPU dependency required.

Caveats

- Does **not** change security or privacy properties.
- Increases binary size and introduces driver-level trust (mitigated by open-source kernels and verifiable computation).
- NVIDIA-specific initially; long-term goal is vendor-agnostic GPU support.
- Power/thermal constraints on embedded or mobile devices may limit always-on acceleration.

Roadmap Integration

Phase	Milestone	Benefit
1+	CUDA-accelerated polynomial & hash kernels	Faster MVP proving on desktops/servers
2+	Integrated Winterfell GPU backend	Mobile-friendly liveness proofs
3+	GPU-optimized recursive composition	Practical high-frequency A2A & invasive BCI use

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11. Open Source, Audit, and Interoperability

11.1 Licensing

The source code shipped in this repository is licensed **AGPL-3.0-only**; the documentation and research specification are **CC BY 4.0**. The protocol must remain open and auditable. Patent license: explicit defensive patent pledge for all contributors.

11.2 Audit Plan

Stage	Auditor Type	Scope
Pre-alpha	Community audit (open source)	

Stage	Auditor Type	Scope
		MembraneEvent schema, circuit logic
Alpha	Independent security firm	TEE integration, side-channel analysis
Beta	Academic cryptographers	zk-STARK soundness, recursive composition
Production	Multi-party audit consortium	Full stack, social recovery, coercion resistance, invasive BCI extension safety, A2A delegation integrity

11.3 Attestation transport (hot path)

The Membrane protocol is **transport-agnostic**. Cryptography, To enrollment, IAC, and fail-closed severance do not require NOSTR, Ethereum, or Arweave. They require a **hot-path attestation bus**: a subject-controlled, signed, append-only event log with fanout to WoT witnesses.

Bus requirements:

Requirement	Rationale
Signed events	Subject and witnesses use Ed25519 (or equivalent)
Hash chain	Each event references `prev_cp_hash` or prior event id
Low latency	Supports cadence Δt (target: <10 ms append on self-hosted bus)
Self-hostable	Subject can run the bus without a chain operator
Metadata only	Raw neural/behavioral payloads never on the bus — STARK public inputs only

Canonical event envelope (`MembraneEvent`):

Field	Description
`type`	e.g. `membrane.cp.liveness`, `membrane.cp.router`, `membrane.cp.bci`, `membrane.iac`, `membrane.alert.degraded`
`subject_pubkey`	Compressed public key (DID or npub encoding allowed)

Field	Description
`prev_cp_hash`	Prior Chain Proof hash
`timestamp`	Hybrid Unix timestamp
`payload`	Type-specific public inputs (Merkle roots, model_id, iac_hash, STARK bytes or URI)
`signature`	Over canonical serialization

Each appended event updates the running **bus Merkle tree** (§5.1); `bus_root` in the next CP commits to the full event sequence. Witnesses subscribe to the bus (or receive events out-of-band), verify STARKs, recompute `bus_root`, and return signatures. The membrane node fails closed if the bus is stale, partitioned, or missing events for active channels.

11.4 Cold anchoring profiles

Cold anchors supply audit durability and third-party timestamps. Fail-closed severance reads the hot bus only.

Profile	Role	Cadence	Examples	Tradeoffs
Hot	Liveness, WoT gossip, severance	Every Δt	Self-hosted append-only log; relay-style pub/sub; MQTT; Hypercore; local file + sync	Low latency; subject-controlled
Warm	Shareable snapshots	Hourly–daily	IPFS / self-hosted object store	Audit copies only
Cold A	Permanent audit trail	Weekly+	Arweave bundle of CP chain	Pay-once storage; slow; public permanence[67]
Cold B	Timestamp / root commit	Daily+	Ethereum L2 calldata with `cp_chain_root`	Gas cost; public chain; not every-Δt[68]
Cold C	Bitcoin time anchor	Daily+	OpenTimestamps over `ots_digest` (§5.1)	Hours to confirm; digest-only on chain; calendar pending until upgrade[69]

Cold C procedure:

1. Build `RollupBundle` with `cp_chain_root` = Merkle root over all `H(CP)` in the period.
2. Sign bundle; compute `ots_digest = SHA256(canonical_json(bundle) || signature)` (same binding as NIP-3B `id+sig` for signed attestations).

3. `ots stamp` → submit to ≥2 calendar pools → `ots upgrade` when Bitcoin confirms.
4. Publish `membrane.anchor.ots` on the hot bus with `{ target: ots_digest, ots_b64, period_end }` so witnesses see intent before Bitcoin confirms.

Design rule: Fail-closed logic reads the hot bus only. Arweave, Ethereum, and Bitcoin OTS do not substitute for a missing router CP or stale witness set. NOSTR relay profile: Appendix B.

11.5 Interoperability

- **TEE ecosystems:** Targets AMD SEV-SNP and Intel TDX via standard attestation APIs. ARM CCA as future target.
- **Identity bridges:** DID:web or DID:ion mapping for W3C decentralized identity interop.
- **ZK ecosystems:** STARK proofs verifiable on StarkNet or Cairo VM; usable for Cold B root commits.
- **Invasive BCI:** Requires proprietary manufacturer API/SDK. If unavailable, the protocol falls back to Liveness-1 without degradation.
- **A2A frameworks:** Can be layered over MCP, A2A, or custom agent communication protocols. CP exchange remains bus-agnostic.

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12. Ethical Considerations

12.1 Digital Minds and Agent Exit Rights

If e-processes or pure agents achieve legal or moral personhood, the membrane must not become a tool of digital imprisonment. The dead-man's key and fail-closed behavior must preserve the right to exit — including the right to sever the attestation chain without operator permission.

12.2 Coercion and Autonomy

The membrane's channel requirements could be weaponized: a captor forces the subject to maintain attested sessions. The duress signal and dead-man's key are partial mitigations, not solutions.

Invasive BCI extension adds new coercion vectors: - Forced implantation or replacement. - Adversarial stimulation to induce compliance. - Neural data extraction under duress. - Mitigation: explicit informed consent requirements, right to refuse implantation, and medical ethics oversight.

Agent coercion: An operator may force an agent to run unauthorized code or delegate to compromised peers. The substrate canary detects code hash mismatch, but if the operator controls the TEE, suppression is possible. Self-hosted TEEs are the primary mitigation.

12.3 Right to Be Forgotten

T_0 anchors are permanent by design. A revocation mechanism must exist that allows an identity to be **explicitly retired** (not recycled) with a final CP signed by the original WoT. This is irreversible and prevents zombie identity attacks.

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13. Known Open Problems

These are explicitly unresolved and invite community contribution. **Top-priority problems are marked with ⭐.**

1. Formal verification of the recursive composition. No complete machine-checked proof exists for the multi-party STARK RFA chain. This is the highest-priority open problem for protocol credibility.

2. High-frequency A2A attestation overhead. zk-STARK generation may be too slow for real-time agent handshakes. Lightweight alternatives (cached proofs + Merkle state diffs) or GPU acceleration (§10.1) need research.

3. Long-term neural template drift and fuzzy extractor stability. Neural patterns change over time due to plasticity, aging, injury, and device degradation. **Closed-loop BCI and wetware sessions additionally shift population statistics on minute scales** during attested use — replay windows and fuzzy-commitment tolerance must account for both session-conditioned and long-term drift.[707172]

1. **Mobile STARK generation.** Current provers are too slow for smartphone-native proof generation. Requires circuit optimization or delegated proving with privacy preservation.

2. **TEE side-channel resistance.** No TEE currently offers provable resistance to all known side-channel attacks. Hardware advances needed.

3. **Coercion-proof authentication.** All liveness signals are vulnerable to coercion. A coercion-proof primitive would require fundamental advances in behavioral biometrics or neurochemistry.

4. **Social graph Sybil resistance at scale.** The WoT model works for small K. Scaling to thousands of witnesses while maintaining recursive validity is an open graph-theoretic problem.

5. **Quantum-resistant hash functions for STARKs.** While hash-based cryptography is post-quantum, specific hash functions used in STARKs (e.g., Rescue, Poseidon) require continued cryptanalysis.

6. **Economic sustainability without tokens.** The no-token incentive model relies on altruism and self-interest. Long-term sustainability of relay and witness networks is unproven.

7. **Agent sovereignty vs. operator ownership.** The protocol proves agent integrity, but does not resolve legal or economic ownership. If an operator owns the hardware, can they claim to own the agent's attestations? This is a governance and legal problem, not a cryptographic one.

☆ **11. Cognitive identity drift detection.** Distinct from neural template drift (problem 3): the attestation chain remains valid while the subject's effective reasoning migrates to external LLM/BCI closed loops. Detecting routing substitution and epistemic capture **without reading thought content** — using IAC scope, router CP consistency, and cross-domain disagreement only — is an open problem tied to §0.2 and §4.2.[73]

☆ **12. Closed-loop plasticity vs. attestation cadence.** How should Δt, `task_id`, and `stimulation_policy` in IACs be chosen when bidirectional channels induce measurable neural drift within minutes?[74] Too-short Δt may false-positive sever; too-long Δt widens the replay window for task-matched neural streams.

13. Observable boundary rules for Liveness-2. Can neural CP verification be expressed as **P-system-style boundary rules** (passage legality of spike-feature objects across the attested membrane) so STARK constraints prove rule compliance without disclosing spike trains?[75][76]

Part 2: zk-STARK Circuit Architecture

Circuit Comparison

Property	Liveness-1	Liveness-2	Liveness-A
Target entity	Biological human	Biological human + BCI	Pure silicon agent
Columns	8	9 (+ `neural_acc`)	7 (+ `ilk_acc`)
Hash accumulator	`sensor_acc`	`sensor_acc` + `neural_acc`	`trace_acc` + `ilk_acc`
Witness layer	Social graph (≥K humans)	Social graph (≥K humans)	Optional operator endorsement
Liveness source	IMU / video / audio	IMU / video / audio + neural features	Execution trace commitments
TEE attestation	node only	node + implant telemetry	Agent enclave attestation
Recursive input	`pub_node_proof_hash`	`pub_node_proof_hash`	`pub_node_proof_hash` + `pub_parent_task_cp`
Proof size	~25 KB	~35 KB	~25 KB

Property	Liveness-1	Liveness-2	Liveness-A
Use case	Channel integrity (human)	BCI-augmented channel integrity	Agent substrate canary
Dependencies	Smartphone / wearable	BCI device (future)	Self-hosted TEE

Liveness-1 Circuit (Traditional Sensors)

Overview

This circuit proves that **some entity** with access to the biometric key and sensor devices generated a liveness signal within a time window, linked to an identity chain, and gathered social attestation. It explicitly does **not** prove consciousness, free will, or absence of coercion.

Property	Value
Circuit Name	`Liveness-1`
Proof System	zk-STARK (hash-based, transparent setup)[77][78]
Hash Function	Poseidon2 (AIR efficiency) + SHA-256 (NIST PQC alignment)[79]
Field	64-bit Goldilocks
Target	Winterfell (Rust) or Cairo VM[80]

Public Inputs

Register	Description
`pub_identity`	Subject public key (compressed; DID or npub encoding)
`pub_prev_cp_hash`	Hash of CP(t-1)
`pub_timestamp`	Hybrid Unix timestamp
`pub_merkle_root`	Merkle root of sensor chunk hashes
`pub_witness_count`	Valid witness signatures (≥ K)
`pub_witness_root`	Merkle root of witness public keys (anonymized)
`pub_node_hash`	Hash of node code + TEE attestation

Private Inputs

Register	Description
`priv_sensor_data`	IMU vectors, video frame hashes, audio fingerprints
`priv_biometric_key`	Derived key from fuzzy commitment of T_0 anchor[81]

Register	Description
`priv_witness_sigs`	Witness signatures on bus event
`priv_location_entropy`	Optional noise hash (omittable)

Execution Trace (AIR)

8 columns, N rows (scales with witness count).

Col	Name	Role
0	`clk`	Clock counter
1	`sensor_acc`	Accumulator for sensor data hash chain
2	`witness_acc`	Accumulator for verified witness signatures
3	`time_delta`	Decrementing: `Δt - clk`
4	`continuity`	Boolean: 1 if `sensor_acc` matches `pub_merkle_root` at final row
5	`sig_valid`	Boolean: 1 if witness signature verifies
6	`threshold`	Boolean: 1 if `witness_acc >= K` at final row
7	`prev_link`	Hash of `pub_prev_cp_hash || pub_identity || pub_timestamp`

Transition Constraints

C1 — Sensor Chain Integrity:

```
sensor_acc' = Hash(sensor_acc || priv_sensor_data[row])
```

Enforces Merkle root is built from provided sensor data. ***Does not prove sensors were honest.***[82]

C2 — Witness Accumulator:

```
witness_acc' = witness_acc + sig_valid
```

C3 — Time Bound:

```
time_delta' = time_delta - 1
time_delta >= 0
```

C4 — Signature Verification:

```
sig_valid = VerifyEd25519(witness_sig, witness_pubkey,
event_hash)
```

C5 — Continuity Link:

```
prev_link = Hash(pub_prev_cp_hash || pub_identity ||
pub_timestamp)
```

Boundary Constraints

First row: `sensor_acc = 0`, `witness_acc = 0`, `time_delta = Δt`, `continuity = 0`

Last row: `sensor_acc == pub_merkle_root`, `witness_acc >= K`, `time_delta >= 0`, `continuity == 1`

Recursive Composition (RFA)

Additional public input:

Register	Description
`pub_node_proof_hash`	Hash of node's own STARK proof from T-1

Last-row constraint:

```
prev_link = Hash(pub_prev_cp_hash || pub_node_proof_hash || pub_identity)
```

This creates a **proof chain**. An adversary compromising the node at T must forge the entire historical chain of node attestations across ≥2/3 independent nodes.[83]

Liveness-2 Neural Extension

Overview

The `Liveness-2` circuit extends `Liveness-1` by adding a **neural feature accumulator** column. It is designed for BCI-equipped users (high-bandwidth cortical implant) and operates as a parallel domain when BCI is registered. The protocol accepts either `Liveness-1` or `Liveness-2` signals, or both for higher confidence.

Formal note: Commitment objects align with **spiking neural P systems** — information carried by spike timing and inter-spike intervals rather than static electrode snapshots.[84] Future circuits may treat `priv_neural_data` rows as observable **boundary events** (passage legality) without revealing spike content, in line with P-system boundary-rule literature.

Additional Public Inputs

Register	Description
`pub_neural_root`	Merkle root of neural feature chunk hashes
`pub_neural_task_id`	Identifier of the standardized neural task performed (e.g., "imagined_fist_5s")

Additional Private Inputs

Register	Description
`priv_neural_data`	Processed neural features: spike rates, LFP power bands, population vectors

Modified Execution Trace (AIR)

The `Liveness-2` trace has **9 columns** (adding column 8):

Col	Name	Role
0–7	*(same as Liveness-1)*	*(same as Liveness-1)*
8	`neural_acc`	Accumulator for neural feature hash chain

Additional Transition Constraints

C6 — Neural Chain Integrity:

```
neural_acc' = Hash(neural_acc || priv_neural_data[row])
```

Enforces neural Merkle root is built from provided neural features. ***Does not prove the neural data came from a conscious brain.***[85]

Modified Boundary Constraints

First row: `neural_acc = 0` (added to existing first-row constraints)

Last row: `neural_acc == pub_neural_root` (added to existing last-row constraints)

Security Properties

- Proves that an entity with the biometric key generated a neural-feature Merkle root within Δt for a specific task.
- Does **not** prove the neural data was not synthesized by a connectome model.
- Does **not** prove the entity was conscious during the task.
- Does **not** prove the implant was not backdoored at manufacture.
- Active challenge-response (future) adds an additional constraint: the neural response must match a stimulation-locked pattern, raising simulation cost.[86]

A2A Agent Circuit (Liveness-A)

Overview

The `Liveness-A` circuit is a simplified variant for pure-silicon agents. It removes the biological/social witness layer and replaces it with substrate

attestation, an Immutable Logging Kernel (ILK) for complete decision audit trails, and operator endorsement when configured.

Public Inputs

Register	Description
pub_agent_identity	Agent public key or DID
pub_prev_cp_hash	Hash of CP(t-1)
pub_timestamp	Hybrid Unix timestamp
pub_code_hash	Hash of agent code / model weights commitment
pub_tee_attestation	TEE remote attestation report
pub_node_hash	Hash of node code + TEE attestation
pub_parent_task_cp	Hash of parent task CP (for delegation chains)
pub_ilk_root	Merkle root of Immutable Logging Kernel decision log

Private Inputs

Register	Description
priv_execution_trace	Commitments to key nonces, decision logs, or state transitions
priv_decision_log	Immutable Logging Kernel (ILK) — Merkle root of all agent decisions since last CP
priv_operator_sig	Optional operator endorsement signature

Execution Trace (AIR)

6 columns, N rows.

Col	Name	Role
0	clk	Clock counter
1	trace_acc	Accumulator for execution trace commitments
2	ilk_acc	Accumulator for Immutable Logging Kernel decision hashes
3	time_delta	Decrementing: Δt - clk
4	continuity	Boolean: 1 if trace_acc matches pub_code_hash at final row
5	tee_valid	Boolean: 1 if TEE attestation verifies
6	prev_link	Hash of pub_prev_cp_hash || pub_agent_identity || pub_timestamp

Transition Constraints

C1 — Execution Trace Integrity:

```
trace_acc' = Hash(trace_acc || priv_execution_trace[row])
```

C2 — Immutable Logging Kernel (ILK):

```
ilk_acc' = Hash(ilk_acc || priv_decision_log[row])
```

Enforces that every agent decision since the last CP is committed into the attestation chain. This creates a complete behavioral audit trail that proves not just code integrity, but decision continuity.

C3 — TEE Attestation Verification:

```
tee_valid = VerifyTEEAttestation(pub_tee_attestation,
pub_code_hash)
```

C4 — Continuity Link:

```
prev_link = Hash(pub_prev_cp_hash || pub_agent_identity ||
pub_timestamp)
```

Boundary Constraints

First row: `trace_acc = 0`, `ilk_acc = 0`, `time_delta = Δt`, `continuity = 0`, `tee_valid = 0`

Last row: `trace_acc == pub_code_hash`, `ilk_acc == pub_ilk_root`, `time_delta >= 0`, `continuity == 1`, `tee_valid == 1`

Sovereignty note

The `Liveness-A` circuit proves **process integrity**, not **goal alignment**. A sovereign agent can prove it is running the code it claims to be running, but this does not prove the code is safe or benevolent. Sovereignty proves runtime integrity, not trustworthiness or alignment.

Integration: Attestation Bus + Local Automation

Hot path (§11.3): Publish `MembraneEvent` records to the self-hosted attestation bus after each proof cycle: `membrane.cp.liveness`, `membrane.cp.router`, `membrane.cp.bci`, `membrane.iac`, `membrane.cp.agent`, `membrane.anchor.ots`.

Cold path (§11.4): Daily `cp_chain_root` rollup; OpenTimestamps stamp on `ots_digest` (Cold C); Arweave bundle (Cold A) or L2 commit (Cold B) on slower cadence.

Local automation (e.g., cron or sensor trigger): - Trigger: IMU threshold or cron. - Action: Compile sensor data → generate STARK → append signed event to bus. - Fallback: Proof failure or missing witnesses → emit `membrane.alert.degraded` → fail closed.

NOSTR profile: Appendix B maps `MembraneEvent` types to kinds `31990`/`31991`.

Citation Tiers

Tier	Type	Citations
T1: Formal	Peer-reviewed or established cryptographic standards	[87] Neulinger & Sparer (Springer), [88] Juels & Wattenberg (ACM CCS), [89] Ben-Sasson et al. (STARKs), [90] NIST PQC, [91] Shamir (ACM), [92] Păun (membrane computing), [93] Ionescu et al. (SN P systems), [94] Gorla et al. (security membranes)
T2: Implementation	Production systems, protocols, hardware docs	[95] AMD SEV-SNP / Intel TDX, [96] Winterfell/Cairo, [97] attestation transport examples, [98] clinical cortical implant literature, [99] DishBrain/CL1 (empirical wetware)
T3: Conceptual	Preprints, heuristics, scope boundaries	[100] Khan preprint (cognitive firewall concept), [101] consciousness out-of-scope, [102] simulation-cost heuristic, [103] BioLLM indie demos

No citation in this document is presented as stronger than its tier.

Appendix A: Version History

Version	Date	Notes
v0.9.16	2026-11-11	Ordered log tree implemented (`membrane-core::log_tree`: membrane-log/v1 construction, signed tree head, inclusion and consistency proofs, RFC 6962 vectors);

Version	Date	Notes
		RollupBundle gains `log_root`/`log_size` and the OTS digest covers them (§5.1.2); rollup bundle schema 0.9.16
v0.9.15	2026-11-11	Split §5.1 into the set commitment tree (§5.1.1, guarantee documented) and the ordered log tree (§5.1.2, RFC 6962-style, versioned domain tag); resolves the leaf-count/order design discussion
v0.9.14	2026-06-19	Păun/Gorla/SN-P lineage (§0.6), BioLLM Class I hybrid, footnotes [104]_[105], Liveness-2 formal note, open problem 13
v0.9.13	2026-06-19	DishBrain/Cortical Labs grounding: closed-loop feedback thesis (§0.2), wetware threat class (A9, Class H), revised §4.6 Liveness-2/stimulation assumptions, IAC fields for `decoder_version`/ `stimulation_policy`, open problem 12
v0.9.12	2026-06-14	Editorial pass: remove redundant required/ optional labels; tighten Merkle and anchoring prose
v0.9.11	2026-06-14	Merkle tree spec (§5.1); OpenTimestamps Cold C; de-sci-fi Gated I/O layer; trim mystical framing
v0.9.10	2026-06-14	Transport-agnostic attestation bus; optional Arweave/L2 cold profiles; NOSTR

Version	Date	Notes
		demoted to appendix example
v0.9.9	2026-06-14	Restore sovereignty terminology; single affiliation disclaimer
v0.9.8	2026-06-14	Replace ceremony/ manifesto phrasing; standardize on local control
v0.9.7	2026-06-14	Remove unrelated-application wording remnants
v0.9.6	2026-06-14	Genericize §0.8 PoP table and invasive BCI (remove named vendors); fix §0.5 affiliation text
v0.9.5	2026-06-14	Remove third-party competitor citations (policy vs continuity §4.8 retained)
v0.9.4	2026-06-14	Independent research / no-affiliation disclaimer (title block, §0.5, contribute)
v0.9.3	2026-06-14	IAC layer (§4.2.1), router session CPs (§4.2.2), epistemic capture (§0.2), substrate parity, policy vs continuity (§4.8), hybrid human–AI membrane, cognitive identity drift open problem
v0.9.2	2026-06-14	Appendix B: open research & Phase 0 stack; published to GitHub
v0.9.1	2026-06-14	Remove cultured tissue (§4.8) and Related Work (§14)
v0.9	2026-06-14	Refocus on cognitive firewall; remove unrelated application-layer conflation
v0.8	2026-06-14	

Version	Date	Notes
		Strip AI slop; pluggable channels; rebrand from GORGONEUM
v0.6	2026-05-12	Related work, GPU section, A2A and invasive BCI extensions
v0.5	2026-05-12	Sovereignty framing and agent attestation mode

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Appendix B: Open Research & Prototype Stack

Clinical cortical implants: what is public

High-bandwidth invasive BCIs have **no open third-party attestation SDK** suitable for Membrane integration today. Useful public material (vendor-neutral):

Resource	URL	Use for Membrane
ASIC / dev platform paper	bioRxiv 703801	Streaming model, electrode counts, UDP multicast — representative implant-class architecture
BCI cybersecurity survey	arXiv: 2007.09466	BLE pairing, firmware updates, companion app as TCB
Wearable BCI security (Argus)	arXiv: 2201.07711	Information-flow control patterns for neural data paths

Treat invasive cortical implants as a **future Liveness-2 target**, not a Phase 0 build dependency.

Closed-loop wetware: DishBrain & commercial precedents

Empirical and commercial closed-loop read/write systems ground Membrane assumptions in §0.2, §4.6, and Class H (outsourced wetware compute). Cited for **channel mechanics only** — not consciousness, sentience adjudication, or third-party attestation SDK availability.

Resource	URL	Use for Membrane
DishBrain (Kagan et al., Neuron 2022)	Neuron · PMC · DOI	**Feedback required for goal-directed plasticity**; open-loop sensation insufficient; minute-scale

Resource	URL	Use for Membrane
		connectivity drift during gameplay; weak between-session retention; HD-MEA multiplexing (many physical electrodes, fewer read/ stim channels)
Cortical Labs (CL1, biological cloud)	corticallabs.com	**Code-deployable wetware compute** — closed-loop API threat without LLM in path; no public membrane attestation SDK
OpenMEA	bioRxiv 2022.11.11.516234	Benchtop closed-loop electrophysiology (lighter-weight precedent than DishBrain)

Engineering takeaways (from DishBrain)

1. **Sever feedback, not only reads.** Unpredictable feedback following "incorrect" motor output drives learning; sensory input without causal consequence does not.
2. **Bind `task_id` + `stimulation_policy` in IAC.** Neural Merkle roots are task- and session-conditioned; replay resistance assumes matched closed-loop history.
3. **Δt must respect minute-scale plasticity.** Measurable electrophysiological change within ~5 minutes of closed-loop embodiment; attestation cadence cannot assume static neural fingerprints.
4. **Do not import "sentience" marketing.** Source paper uses a narrow active-inference definition; Cortical Labs consumer copy escalates further. Membrane attests channels, not phenomenology.

Formal membrane lineage (Păun, Gorla, SN P)

The Membrane name overlaps with **membrane computing** (Păun 1998) and **security membranes** (Gorla et al.). See whitepaper §0.6 for the four-way distinction. These are **formal and security precedents** — not claims that Cortical Labs runs P-system simulators.

Resource	URL	Use for Membrane
Păun (2000) — *Computing with Membranes*	DOI · Scholarpedia overview	P systems: compartments, symport/ antiport, dissolution → IAC passage rules, severance
Păun & Rozenberg (2002) — symport/ antiport	New Generation Computing	Paired in/out transport → bidirectional BCI `stimulation_policy`
Ionescu, Păun, Yokomori (2006) — SN P systems	Fundamenta Informaticae · ENTCS	Spike timing as object type → Liveness-2 commitment design

Resource	URL	Use for Membrane
Gorla, Hennessy, Sassone (2002) — security membranes		Site filter on agent/code entry → membrane-as-firewall precedent

P-system → Membrane primitive map

P system	Membrane
Skin / environment	Endogenous vs exogenous cognition
Symport / antiport	IAC-gated read/write channels
Promoter / inhibitor	IAC scope, TTL, allowlists
Membrane dissolution	Fail-closed severance
SN P spike trains	Liveness-2 `priv_neural_data` rows

Do not conflate: Păun's field studies abstract cell-inspired computation. Cortical Labs ships **physical** closed-loop electrophysiology. The Membrane attests **which couplings are authorized**.

Merkle trees

All bulk commitments use **SHA-256 Merkle trees** per whitepaper §5.1:

Tree	Built from	When
Channel	`H(0x00 ‖ feature_chunk)` from EEG/IMU/router context chunks	Every CP / Δt
Bus	`H(0x01 ‖ canonical MembraneEvent)`	Every bus append
Witness	`H(0x03 ‖ witness_pubkey)`	Every CP with WoT
CP chain rollup	`H(0x02 ‖ cp_hash)`	Daily cold anchor input

Implement with a single `merkle_sha256(leaves: bytes[], domain: u8)` helper; sort leaves lexicographically before pairing.

Attestation transport profiles (Phase 0)

The protocol is **transport-agnostic**. Pick a hot bus for every-Δt CPs; add cold anchors for audit timestamps and durability.

Profile	When	Implementation sketch
Hot	Every Δt	Self-hosted append-only log + bus Merkle root in each CP
Warm	Hourly+	IPFS or object store for STARK bundles + signed rollup JSON
Cold A	Weekly+	Arweave bundle containing rollup + `.ots`

Profile	When	Implementation sketch
Cold B	Daily+	L2 calldata commit of `cp_chain_root`
Cold C	Daily+	OpenTimestamps on `ots_digest` (§5.1)

Cold C: OpenTimestamps rollup

```
# 1. Export daily rollup
membrane export-rollup --day 2026-06-14 --out rollup.json
# rollup.json contains cp_chain_root, last_bus_root,
        subject_pubkey, period_*

# 2. Sign canonical JSON → compute ots_digest = SHA256(json ||
        sig)
membrane sign-rollup rollup.json --out rollup.signed.json

# 3. Stamp (submit to ≥2 calendars)
ots stamp rollup.signed.json
# → rollup.signed.json.ots (pending)

# 4. After Bitcoin confirms (hours later)
ots upgrade rollup.signed.json.ots
ots verify rollup.signed.json.ots

# 5. Announce on hot bus before Bitcoin confirms
# MembraneEvent type membrane.anchor.ots { target: ots_digest,
        ots_b64, period_end }
```

Tools: `pip install opentimestamps-client`; Bitcoin headers via local `bitcoind` (pruned OK) or Esplora.

Binding rule: Stamp `SHA256(canonical_bundle || signature)`, not `cp_chain_root` alone — same rationale as NIP-3B `id+sig` (prevents backdated signature replay).

Example hot bus: NOSTR relay profile

One practical Phase 0 backend maps `MembraneEvent` to NOSTR application events:

`MembraneEvent.type`	NOSTR mapping
`membrane.cp.liveness`	kind `31990`, tag `["k", "the-membrane-liveness"]`
`membrane.cp.router`	kind `31990`, tag `["k", "the-membrane-router"]`
`membrane.cp.bci`	kind `31990`, tag `["k", "the-membrane-bci"]`
`membrane.iac`	kind `31990`, tag `["k", "the-membrane-iac"]`
`membrane.anchor.ots`	

`MembraneEvent.type`	**NOSTR mapping**
	kind `31990`, tag `["k", "the-membrane-anchor-ots"]`
`membrane.alert.degraded`	kind `31991`

Common tags: `["e", <prev_event_id>]`, `["p", <subject_pubkey>]`.
Run a **self-hosted relay**; treat public relays as optional read replicas only.

Other hot-bus options: MQTT broker, Hypercore feed, SQLite append log + Tailscale sync.

Open acquisition & decode (Phase 0 hardware)

Project	License	Role
BrainFlow	MIT	Unified SDK for OpenBCI, Muse, etc. — channel registration + feature extraction
OpenBCI	Open hardware + OSS tooling	Lab bench; Galea adds EEG/EMG/PPG/EDA in XR
Lab Streaming Layer	BSD	Time-synced multimodal bus — use as **membrane data plane**
MetaBCI	GPL-2.0	Full decode pipeline; study **decoder drift** when models retrain
PyNoetic	GPL-3.0	Real-time EEG → classifier pipelines

Invasive / bench research (not consumer implants)

Project	Notes
Iris-128	Open 128-ch headstage (CERN-OHL)
OpenMEA	Benchtop closed-loop electrophysiology
DishBrain / MaxOne HD-MEA	See Closed-loop wetware — strongest empirical closed-loop precedent cited in v0.9.13

Security & firewall research (closest to Membrane thesis)

Project	Type	Membrane mapping
iba-neural-guard	Open repo	**Decoded signal ≠ authorization.** Signed intent certificate before any BCI action; blocks capability drift
Argus / LibArgus	Paper (no public repo)	OS **information flow control** for wearable BCIs; 300+ vulns on Muse/ NeuroSky/OpenBCI stacks
NSP — Neural Sensory Protocol	Spec draft	PQC wire protocol for neural frames; constant-rate BLE against timing side channels
NeuroZKP	Project site	ZK proofs over neural signals — prove authorization without raw spike disclosure

iba-neural-guard is the closest open articulation of the policy layer Membrane needs: the thought is not the authorization; the signed certificate is.

LLM ↔ brain coupling (adversary models)

Active research on **closed-loop brain → LLM routing** — where identity drift and sequestration appear first:

Project	URL	Why it matters
NeuroLM	github.com/935963004/NeuroLM	EEG treated as "foreign language" inside an LLM — native cognition absorbed into model space
SYNAPTICON	github.com/AlbertBarqueDuran/SYNAPTICON	EEG → text → LLM → output closed loop
Brain-LLM Interface	arXiv:2603.16897	EEG gates LLM refinement at inference time
BioLLM (CL1 + LLM)	GitHub: 4R7I5T/CL1_LLM_Encoder	**Class I:** wetware re-weights token logits inside router loop — requires linked `membrane.cp.router` + `membrane.cp.bci`

Wetware closed-loop (no LLM in path)

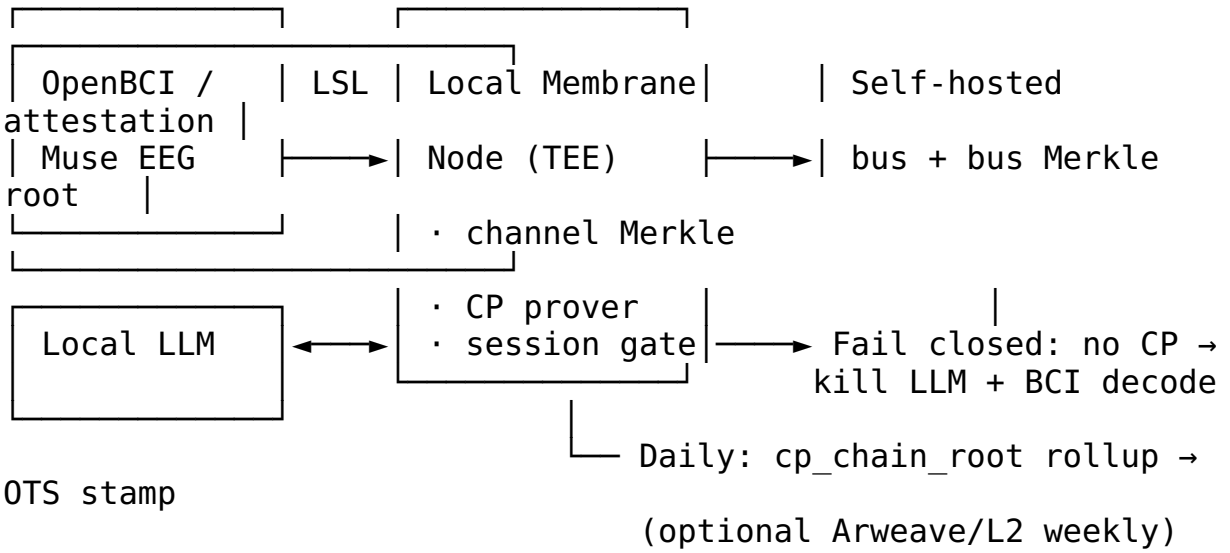
Project	URL	Why it matters
Cortical Labs CL1	corticallabs.com	

Project	URL	Why it matters
		Local code-deployable biological computer — cognition routed through external neurons
Cortical Cloud	corticallabs.com	Remote biological compute API — substrate-transition / Class H threat
Bio-datacenter prototypes (2026)	NUS Medicine · DCD	Melbourne (~120 CL1) + Singapore (phased); wetware at datacenter scale
DishBrain (research)	Neuron 2022	Defines minimal closed-loop mechanics: read → act → feedback required for plasticity

Membrane term	Meaning in this stack
Identity drift	Decoder or LLM updates, session fork, model swap without new Chain Proof
Sequestration	Cognition lives in external inference (LLM **or wetware**, including **BioLLM hybrids**); endogenous loop bypassed or unreadable
Firewall job	Gate which channels may couple; sever on missing/stale CP; **sever feedback** when bidirectional policy breaks

---

Suggested Phase 0 architecture



Build order

1. **BrainFlow + LSL** — stream EEG features; Merkle-commit chunks locally; never publish raw traces.
2. **Channel registry** — YAML list of permitted paths (BCI app, local LLM port, wetware API endpoints, forbidden cloud URLs).
3. **Merkle + Winterfell Liveness-1** — prove `pub_merkle_root` over feature vectors + timestamp (§Part 2).

4. **Intent gate** — adapt iba-neural-guard: no signed scope → no decode→action mapping.
5. **Attestation bus** — append signed `MembraneEvent`; maintain bus Merkle tree; NOSTR relay profile if used.
6. **Daily OTS rollup** — sign `RollupBundle`, `ots stamp`, `ots upgrade` when confirmed.
7. **WoT** — K=2 human witnesses sign CP validity out-of-band (Signal/video).

What Phase 0 proves

- Subject can run membrane-gated channels **without default cloud LLM routing**.
- Unauthorized channel open → detectable in attestation chain (if channel touches the bus).
- Daily `cp_chain_root` + OTS gives third-party **existence time** for the rollup (audit, not liveness).
- Subject can **sever** all links (dead-man's key / kill switch).

What Phase 0 does not prove

- Covert passive surveillance with no attested channel.
- Thought content or consciousness.
- Invasive cortical implant integration.

References to add to implementation issues

- Păun, G. (2000). *Computing with Membranes*. J. Comput. System Sci. https://doi.org/10.1006/jcss.2000.1773 — P-system formal lineage (§0.6).
- Ionescu, M., Păun, G., & Yokomori, T. (2006). *Spiking neural P systems*. Fundamenta Informaticae — Liveness-2 spike objects (§Part 2).
- Gorla, D., Hennessy, M., & Sassone, V. (2002). *Security Policies as Membranes*. ENTCS — security-boundary precedent (§0.6).
- BioLLM / CL1_LLM_Encoder demos (2026) — Class I hybrid threat; GitHub `4R7I5T/CL1_LLM_Encoder`.
- Kagan, B. J. et al. (2022). *In vitro neurons learn and exhibit sentience when embodied in a simulated game-world*. Neuron. https://doi.org/10.1016/j.neuron.2022.09.001 — closed-loop feedback mechanics; minute-scale plasticity (§0.2, §4.6).
- Cortical Labs. CL1 / Cortical Cloud product pages. https://corticallabs.com — commercial wetware closed-loop precedents (Class H); descriptive only.
- Khan, E. (2026). *Brain Hacking: AI for Safeguarding Against Dangerous AI*. Preprints.org — cognitive firewall framing (T3).
- Li et al. BCI cybersecurity survey — arXiv:2007.09466 — implant companion-app threat model.
- Argus wearable BCI security — arXiv:2201.07711.
- OpenTimestamps — https://opentimestamps.org/ — Bitcoin batched timestamps for rollup digests.

How to Contribute

Contributors are welcome.

Implement the MVP (§9) using the stacks in Appendix B. Publish benchmarks via GitHub issues. Security disclosures: notify the author privately, 90-day coordinated disclosure. The shipped source code is AGPL-3.0-only; the documentation and research specification are CC BY 4.0.

Footnotes

Tags: #the-membrane #whitepaper #zk-stark #identity-attestation #process-continuity #sovereignty #agent-attestation

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1. Păun, G. (2000). *Computing with Membranes*. Journal of Computer and System Sciences 61(1), 108–143. https://doi.org/10.1006/jcss.2000.1773 (TUCS Report 208, 1998). — **Formal (T1).** Foundational P systems: nested membranes, symport/antiport transport, dissolution. Formal precedent for IAC-gated channel passage — **not** a claim that Cortical Labs implements P-system semantics. Overview: http://www.scholarpedia.org/article/Membrane_Computing↩

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4. NeuroLM, SYNAPTICON, Brain-LLM Interface — see Appendix B. **Conceptual / active research (T3).** Closed-loop EEG→LLM pipelines illustrate sequestration and identity drift threats; not production security claims.↩

5. Kagan, B. J. et al. (2022). *In vitro neurons learn and exhibit sentience when embodied in a simulated game-world*. Neuron 110(23), 3952–3969. https://doi.org/10.1016/j.neuron.2022.09.001 — **Peer-reviewed empirical (T1 for closed-loop mechanics; T3 for sentience framing).** DishBrain HD-MEA closed-loop read/write: learning requires

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7. Gödel, K. (1931). *Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I*. — **Formal limit (T1).** Any self-referential proof system faces consistency limits. Cross-domain consensus is the practical workaround, not a solution.↩

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12. Ionescu, M., Păun, G., & Yokomori, T. (2006). *Spiking neural P systems*. Fundamenta Informaticae 71(2–3), 279–308. — **Formal (T1).** Spike-timing as computational object; motivates Liveness-2 spike-rate / interval commitments over raw electrode dumps.↩

13. Gorla, D., Hennessy, M., & Sassone, V. (2002). *Security Policies as Membranes in Systems for Global Computing*. ENTCS 63. https://

doi.org/10.1016/S1571-0661(05)05109-1 — **Formal (T1).** Membranes as site filters controlling external agent actions — independent security-boundary precedent for fail-closed channel policy.↩

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17. National Institute of Standards and Technology (NIST). *SP 800-207: Zero Trust Architecture*. https://doi.org/10.6028/NIST.SP.800-207 — **Standards (T1).** Vendor-neutral reference for Zero Trust Architecture. The Membrane implements these principles via cryptographic consensus rather than network segmentation.↩

18. Neulinger, J. & Sparer, K. (2025). *Fostering AI alignment through blockchain, proof of personhood and zero knowledge proofs*. Cluster Computing. https://link.springer.com/article/10.1007/s10586-025-05729-8 — **Formal / peer-reviewed (T1).** Establishes PoP consensus, zk-STARK identity verification, AI Shield model. Core engineering reference.↩

19. Clinical cortical implant literature (e.g., high-density electrode ASIC platforms, investigational device trials). **Implementation / medical device (T2).** As of 2026, consumer-accessible third-party attestation APIs for invasive BCIs are not publicly available. Thread/lead retraction and firmware rollback remain documented clinical risks. All Membrane integration claims are prospective.↩

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28. Juels, A. & Wattenberg, M. (1999). *A Fuzzy Commitment Scheme*. ACM CCS. — **Formal (T1).** Allows key derivation from noisy biometrics without storing raw templates.↩

29. Grokipaedia. *iba-neural-guard*. https://github.com/Grokipaedia/iba-neural-guard — **Implementation reference (T2).** Intent certificate pattern: decoded neural signal ≠ authorization. Adapted for Membrane IAC layer (§4.2.1).↩

30. Kagan, B. J. et al. (2022). *In vitro neurons learn and exhibit sentience when embodied in a simulated game-world*. Neuron 110(23), 3952–3969. https://doi.org/10.1016/j.neuron.2022.09.001 — **Peer-reviewed empirical (T1 for closed-loop mechanics; T3 for sentience framing).** DishBrain HD-MEA closed-loop read/write: learning requires feedback, not sensation alone; minute-scale plasticity; weak between-session retention. Cortical Labs CL1 / biological cloud cited descriptively as commercial wetware closed-loop precedents — not Membrane endorsements. “Sentience” in source paper follows narrow active-inference usage; Membrane does not adjudicate consciousness.↩

31. Zimmermann, P. (1995). *PGP User’s Guide*. — **Implementation reference (T2).** Web-of-trust model adapted for biological witness attestation.↩

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36. Lamport, L., Shostak, R., & Pease, M. (1982). *The Byzantine Generals Problem*. ACM TOPLAS. — **Formal (T1).** Establishes the 2/3 consensus bound. No recursive mechanism can exceed this threshold without additional trust assumptions.↩

37. Kagan, B. J. et al. (2022). *In vitro neurons learn and exhibit sentience when embodied in a simulated game-world*. Neuron 110(23), 3952–3969. https://doi.org/10.1016/j.neuron.2022.09.001 — **Peer-reviewed empirical (T1 for closed-loop mechanics; T3 for sentience framing).** DishBrain HD-MEA closed-loop read/write: learning requires feedback, not sensation alone; minute-scale plasticity; weak between-session retention. Cortical Labs CL1 / biological cloud cited descriptively as commercial wetware closed-loop precedents — not Membrane endorsements. “Sentience” in source paper follows narrow active-inference usage; Membrane does not adjudicate consciousness.↩

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54. Ben-Sasson, E. et al. (2018). *Scalable, transparent, and post-quantum secure computational integrity*. IACR ePrint. — **Formal (T1).** Foundational STARK paper. Winterfell (Rust) and Cairo are production implementations.↩

55. Shamir, A. (1979). *How to Share a Secret*. Communications of the ACM. — **Formal (T1).** Secret sharing scheme used for T_0 key sharding and social recovery.↩

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63. Zimmermann, P. (1995). *PGP User's Guide*. — **Implementation reference (T2).** Web-of-trust model adapted for biological witness attestation.↩

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67. Arweave permaweb. https://www.arweave.org/ — **Implementation (T2).** Cold A storage for CP chain snapshots. Unsuitable as hot-path liveness bus (latency, pub/sub model).↩

68. Ethereum / L2 rollups. — **Implementation (T2).** Cold B root commits (`H(CP_root)`). Unsuitable for per-Δt CP publication (cost, latency, public mempool).↩

69. OpenTimestamps. https://opentimestamps.org/ — **Implementation (T2).** Batched Bitcoin timestamp proofs. Stamp `ots_digest` from signed rollup bundle; verify with `opentimestamps-client` + Bitcoin headers. Not a liveness bus.↩

70. Juels, A. & Wattenberg, M. (1999). *A Fuzzy Commitment Scheme*. ACM CCS. — **Formal (T1).** Allows key derivation from noisy biometrics without storing raw templates.↩

71. Clinical cortical implant literature (e.g., high-density electrode ASIC platforms, investigational device trials). **Implementation / medical device (T2).** As of 2026, consumer-accessible third-party attestation APIs for invasive BCIs are not publicly available. Thread/lead retraction and firmware rollback remain documented clinical risks. All Membrane integration claims are prospective.↩

72. Kagan, B. J. et al. (2022). *In vitro neurons learn and exhibit sentience when embodied in a simulated game-world*. Neuron 110(23), 3952–3969. https://doi.org/10.1016/j.neuron.2022.09.001 — **Peer-reviewed empirical (T1 for closed-loop mechanics; T3 for sentience framing).** DishBrain HD-MEA closed-loop read/write: learning requires feedback, not sensation alone; minute-scale plasticity; weak between-session retention. Cortical Labs CL1 / biological cloud cited

descriptively as commercial wetware closed-loop precedents — not Membrane endorsements. “Sentience” in source paper follows narrow active-inference usage; Membrane does not adjudicate consciousness.↩

73. NeuroLM, SYNAPTICON, Brain-LLM Interface — see Appendix B. **Conceptual / active research (T3).** Closed-loop EEG→LLM pipelines illustrate sequestration and identity drift threats; not production security claims.↩

74. Kagan, B. J. et al. (2022). *In vitro neurons learn and exhibit sentience when embodied in a simulated game-world*. Neuron 110(23), 3952–3969. https://doi.org/10.1016/j.neuron.2022.09.001 — **Peer-reviewed empirical (T1 for closed-loop mechanics; T3 for sentience framing).** DishBrain HD-MEA closed-loop read/write: learning requires feedback, not sensation alone; minute-scale plasticity; weak between-session retention. Cortical Labs CL1 / biological cloud cited descriptively as commercial wetware closed-loop precedents — not Membrane endorsements. “Sentience” in source paper follows narrow active-inference usage; Membrane does not adjudicate consciousness.↩

75. Păun, G. (2000). *Computing with Membranes*. Journal of Computer and System Sciences 61(1), 108–143. https://doi.org/10.1006/jcss.2000.1773 (TUCS Report 208, 1998). — **Formal (T1).** Foundational P systems: nested membranes, symport/antiport transport, dissolution. Formal precedent for IAC-gated channel passage — **not** a claim that Cortical Labs implements P-system semantics. Overview: http://www.scholarpedia.org/article/Membrane_Computing↩

76. Ionescu, M., Păun, G., & Yokomori, T. (2006). *Spiking neural P systems*. Fundamenta Informaticae 71(2–3), 279–308. — **Formal (T1).** Spike-timing as computational object; motivates Liveness-2 spike-rate / interval commitments over raw electrode dumps.↩

77. Neulinger, J. & Sparer, K. (2025). *Fostering AI alignment through blockchain, proof of personhood and zero knowledge proofs*. Cluster Computing. https://link.springer.com/article/10.1007/s10586-025-05729-8 — **Formal / peer-reviewed (T1).** Establishes PoP consensus, zk-STARK identity verification, AI Shield model. Core engineering reference.↩

78. Ben-Sasson, E. et al. (2018). *Scalable, transparent, and post-quantum secure computational integrity*. IACR ePrint. — **Formal (T1).** Foundational STARK paper. Winterfell (Rust) and Cairo are production implementations.↩

79. National Institute of Standards and Technology (NIST). *Post-Quantum Cryptography Standardization*. — **Standards (T1).** SPHINCS+ selected as hash-based post-quantum signature. Shares hash-function assumptions with zk-STARKs.↩

80. Ben-Sasson, E. et al. (2018). *Scalable, transparent, and post-quantum secure computational integrity*. IACR ePrint. — **Formal (T1).**

Foundational STARK paper. Winterfell (Rust) and Cairo are production implementations.↩

81. Juels, A. & Wattenberg, M. (1999). *A Fuzzy Commitment Scheme*. ACM CCS. — **Formal (T1).** Allows key derivation from noisy biometrics without storing raw templates.↩

82. Heuristic argument. — **Conceptual / speculative (T3).** The claim that physical-sensor signals are “expensive to simulate” is a cost argument, not a cryptographic proof. Vulnerable to replay, injection, and coercion.↩

83. Lamport, L., Shostak, R., & Pease, M. (1982). *The Byzantine Generals Problem*. ACM TOPLAS. — **Formal (T1).** Establishes the 2/3 consensus bound. No recursive mechanism can exceed this threshold without additional trust assumptions.↩

84. Ionescu, M., Păun, G., & Yokomori, T. (2006). *Spiking neural P systems*. Fundamenta Informaticae 71(2–3), 279–308. — **Formal (T1).** Spike-timing as computational object; motivates Liveness-2 spike-rate / interval commitments over raw electrode dumps.↩

85. Clinical cortical implant literature (e.g., high-density electrode ASIC platforms, investigational device trials). **Implementation / medical device (T2).** As of 2026, consumer-accessible third-party attestation APIs for invasive BCIs are not publicly available. Thread/lead retraction and firmware rollback remain documented clinical risks. All Membrane integration claims are prospective.↩

86. Clinical cortical implant literature (e.g., high-density electrode ASIC platforms, investigational device trials). **Implementation / medical device (T2).** As of 2026, consumer-accessible third-party attestation APIs for invasive BCIs are not publicly available. Thread/lead retraction and firmware rollback remain documented clinical risks. All Membrane integration claims are prospective.↩

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88. Juels, A. & Wattenberg, M. (1999). *A Fuzzy Commitment Scheme*. ACM CCS. — **Formal (T1).** Allows key derivation from noisy biometrics without storing raw templates.↩

89. Ben-Sasson, E. et al. (2018). *Scalable, transparent, and post-quantum secure computational integrity*. IACR ePrint. — **Formal (T1).** Foundational STARK paper. Winterfell (Rust) and Cairo are production implementations.↩

90. National Institute of Standards and Technology (NIST). *Post-Quantum Cryptography Standardization*. — **Standards (T1).** SPHINCS+ selected as hash-based post-quantum signature. Shares hash-function assumptions with zk-STARKs.↩

91. Shamir, A. (1979). *How to Share a Secret*. Communications of the ACM. — **Formal (T1).** Secret sharing scheme used for T_0 key sharding and social recovery.↩

92. Păun, G. (2000). *Computing with Membranes*. Journal of Computer and System Sciences 61(1), 108–143. https://doi.org/10.1006/jcss.2000.1773 (TUCS Report 208, 1998). — **Formal (T1).** Foundational P systems: nested membranes, symport/antiport transport, dissolution. Formal precedent for IAC-gated channel passage — **not** a claim that Cortical Labs implements P-system semantics. Overview: http://www.scholarpedia.org/article/Membrane_Computing↩

93. Ionescu, M., Păun, G., & Yokomori, T. (2006). *Spiking neural P systems*. Fundamenta Informaticae 71(2–3), 279–308. — **Formal (T1).** Spike-timing as computational object; motivates Liveness-2 spike-rate / interval commitments over raw electrode dumps.↩

94. Gorla, D., Hennessy, M., & Sassone, V. (2002). *Security Policies as Membranes in Systems for Global Computing*. ENTCS 63. https://doi.org/10.1016/S1571-0661(05)05109-1 — **Formal (T1).** Membranes as site filters controlling external agent actions — independent security-boundary precedent for fail-closed channel policy.↩

95. AMD. *SEV-SNP: Strengthening VM Isolation with Integrity Protection and More*. White paper. Intel. *Trust Domain Extensions (TDX) Architecture Specification*. — **Implementation / hardware docs (T2).** TEEs provide best-available isolation but are vulnerable to side-channels and supply-chain attacks.↩

96. Ben-Sasson, E. et al. (2018). *Scalable, transparent, and post-quantum secure computational integrity*. IACR ePrint. — **Formal (T1).** Foundational STARK paper. Winterfell (Rust) and Cairo are production implementations.↩

97. Attestation transport examples (pub/sub meshes, append-only logs). NOSTR NIPs: https://github.com/nostr-protocol/nips — **Implementation (T2).** Example hot-bus profile; not normative.↩

98. Clinical cortical implant literature (e.g., high-density electrode ASIC platforms, investigational device trials). **Implementation / medical device (T2).** As of 2026, consumer-accessible third-party attestation APIs for invasive BCIs are not publicly available. Thread/lead retraction and firmware rollback remain documented clinical risks. All Membrane integration claims are prospective.↩

99. Kagan, B. J. et al. (2022). *In vitro neurons learn and exhibit sentience when embodied in a simulated game-world*. Neuron 110(23), 3952–3969. https://doi.org/10.1016/j.neuron.2022.09.001 — **Peer-reviewed**

empirical (T1 for closed-loop mechanics; T3 for sentience framing). DishBrain HD-MEA closed-loop read/write: learning requires feedback, not sensation alone; minute-scale plasticity; weak between-session retention. Cortical Labs CL1 / biological cloud cited descriptively as commercial wetware closed-loop precedents — not Membrane endorsements. “Sentience” in source paper follows narrow active-inference usage; Membrane does not adjudicate consciousness.↩

100. Khan, E. (2026). *Brain Hacking: Artificial Intelligence for Safeguarding Against Dangerous AI*. MDPI Preprints. https://www.preprints.org/manuscript/202601.0156 — **Conceptual / preprint (T3).** Proposes “cognitive firewall” as neural anomaly detection. Not peer-reviewed. Framing reference only.↩

101. Chalmers, D. J. (1995). *Facing Up to the Problem of Consciousness*. Journal of Consciousness Studies. — **Philosophical baseline (T3).** Consciousness is not formally represented in this protocol; channel attestation only.↩

102. Heuristic argument. — **Conceptual / speculative (T3).** The claim that physical-sensor signals are “expensive to simulate” is a cost argument, not a cryptographic proof. Vulnerable to replay, injection, and coercion.↩

103. BioLLM-class hybrids (2026): independent developer demos coupling Cortical Cloud CL1 to small local LLMs for token re-weighting (e.g., `4R7I5T/CL1_LLM_Encoder` on GitHub; reported in tech press March 2026). — **Conceptual / active research (T3).** Illustrates Class I (LLM + wetware in one inference loop); not a production security claim or Cortical Labs product specification.↩

104. Păun, G. (2000). *Computing with Membranes*. Journal of Computer and System Sciences 61(1), 108–143. https://doi.org/10.1006/jcss.2000.1773 (TUCS Report 208, 1998). — **Formal (T1).** Foundational P systems: nested membranes, symport/antiport transport, dissolution. Formal precedent for IAC-gated channel passage — **not** a claim that Cortical Labs implements P-system semantics. Overview: http://www.scholarpedia.org/article/Membrane_Computing↩

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